
PHYS 358

Thermal Physics

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Credit to Dr. Adrian Lupascu for His Lectures on PHYS 358

Preface

A majority of the information presented in this note package is transcribed directly from the *PHYS 358: Thermal Physics* lectures during the Fall 2025 term, taught by Dr. Adrian Lupascu.

These notes are by no means exhaustive nor are they guaranteed to be error-free, and I do not make any claims about the reliability or accuracy of these set of notes. They are intended primarily for personal use and for those who may benefit from an alternative presentation of the material.

Please use these notes at your own discretion.

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1 Temperature and Thermal Equilibrium

1.1 Introduction

Consider the following setup:

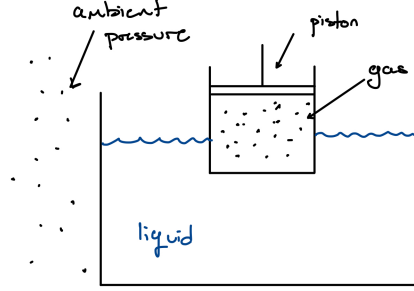


Figure 1: Gas within an open system under ambient pressure.

a gas in a cylinder with a piston is brought into contact with liquid in a recipient. Common sense tells us that, if the gas is temperature, T_g , than the liquid, T_l , then $T_g \uparrow$ as pressure remains constant, as determined by an ambient environment. That is, the gas expands to some finite volume difference.

The governing equation of this system is the **ideal gas law**,

$$PV = kNT \tag{1}$$

where P is a *constant* pressure, V is the volume of the gas, k is the Boltzmann constant, N is the number of particles within the gas, and T is the temperature of the gas. There are a couple observations we can make about this system.

1. The system is *generic*.
2. The size or rate of change of our quantities can depend on the size of the systems.
3. The saturation of the quantities we are interested in.
4. The equilibrium, once set, is maintained.

All of these observations has a common property: *temperature*. Further, we can infer two properties about temperature with just this simple system:

1. After some time, t' , the temperature is the same for two objects under equilibrium.
2. The closeness of temperatures between two objects can be associated with the size of changes in properties of physical systems as they reach equilibrium.

In both properties, we say that the **relaxation time** is the time it takes for a physical system to reach thermal equilibrium. This relaxation time can depend on relative sizes of our quantities involved as they interact. In fact, our quantities exchange *energy*, and the transfer of energy is determined by **heat**.

1.1.1 Heat Transfer Mechanisms

We can have thermal transfer in a system under **conduction**. In particular, we write conduction to occur on a microscopic scale (either by collisions, vibrations, etc.). Two other methods of thermal transfer are **radiation** and **convection**. The more important is convection, and occurs by the mixing of two systems at a microscopic scale.

Now consider two systems under the same experiment with differing initial conditions. We denote system I with the two quantities A and B, same with system II. Assume that system II has one more quantity of B than A.

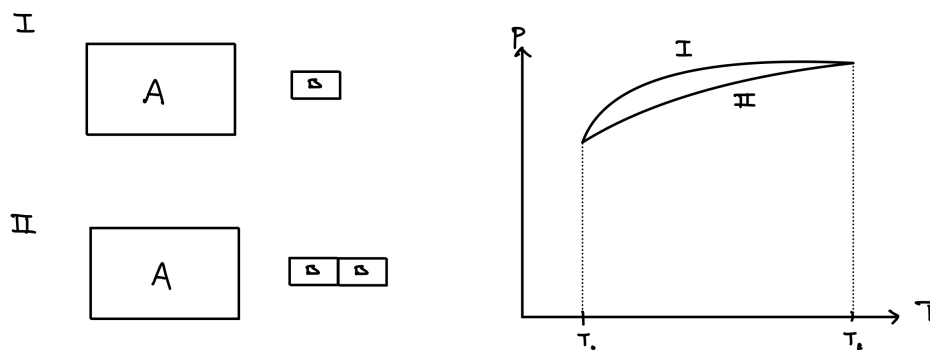


Figure 2: Two systems under the same experiment in different initial conditions. (I) System A and B. (II) System A and twice B.

Notice that for setup II, the system reaches thermal equilibrium faster than setup I (the p vs. T curve becomes constant at the time T_f). Why is this the case? We have that setup II has twice the amount of B that is in setup I. In turn, the energy transfer is faster since there is more mass to transfer energy. Thus, the II curve has less of a gradient than the I curve, and thereby less arc-length implying a faster energy transfer.

1.1.2 Thermodynamic Equilibrium

Equilibrium is not only limited to energy exchange between two objects. We can, in fact, develop the notion of equilibrium with a single object, *homogeneous* from the POV of its constitution, but *inhomogeneous* in regards to its thermodynamic parameters.

We also have other types of equilibrium, not just thermal equilibrium arising as a result of differing initial temperatures of two (or a single) object(s). Table 1 shows the three major types of equilibrium we will encounter.

Type	Quantity Exchanged	Equilibrium Quantity
Thermal	E	T
Mechanical	V	P
Diffusive	Particles	Chemical Potential.

Table 1: Types of equilibrium in thermodynamics.

Here, E is energy; the other variables are as before. It should be noted that while differences in pressure and chemical potential arise from changes in volume and number of particles, we can also associate them with change in energy. Temperature is unique in that it is a parameter that induces energy exchange *independent* to volume or particle number change.

The operational definition of temperature is based on a measurement with a thermometer. Thermometers operate on thermal expansion: $T \uparrow \implies V \uparrow$ since $T \propto V$. We use this to define a temperature scale based on two temperature to two well-defined conditions for a substance.

For a gas thermometer, the change in volume also follows $\Delta V \propto \Delta T$. So, in the limit that $V \rightarrow 0$, we arrive at $T_0 = -273.15^\circ\text{C}$. We call this the **absolute zero**.

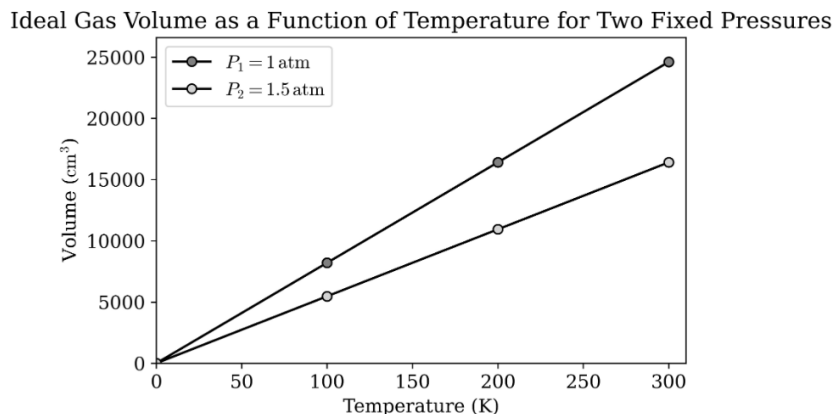


Figure 3: Example data set for measurements of an ideal gas with a constant pressure thermometer for two constant fixed pressures.

Here, notice that T is in Kelvin, so in this case $T \rightarrow 0 \implies V \rightarrow 0$. We can, in fact, use this to infer a conversion from Kelvin to Celsius or vice versa.

1.2 The Ideal Gas

Suppose we have an ideal gas at low density and high temperature. It follows that we can model this gas by the law in (1), the **ideal gas law**. Alternatively, another representation is

$$PV = nRT. \quad (2)$$

In this form, the quantity n is such that

$$n = \frac{N}{N_A} \quad (3)$$

where $N_A := 6.023 \cdot 10^{23}$ is Avogadro's number. We say that n is the number of **moles** of the gas. On the other hand, $k := 1.381 \cdot 10^{-23} \text{ J/K}$ is the **Boltzmann constant**, and together we define

$$R := kN_A = 8.31 \text{ Jmol}^{-1}\text{K}^{-1}.$$

1.2.1 Microscopic Pressure Model

The ideal gas model is based on the approximation that molecules have *no volume* and do *not* interact with each other. Fig. 4 does not provide a model for energy transfer between molecules, but is sufficient to show how pressure relates to microscopic properties.

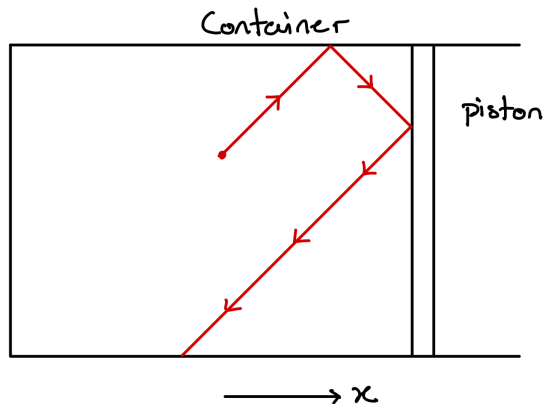


Figure 4: Trajectory of a single gas molecule (red) in a container with a static piston.

We can associate the collision of a molecule with the immobile piston with a change in momenta

$$\Delta p_x = -2mv_x.$$

This, of course, implies a force imparted by the piston given by

$$\langle \mathbf{F}_{\text{piston}} \rangle = \frac{\Delta p_x}{\Delta t} \hat{x} = \frac{\Delta p_x}{2L/v_x} \hat{x} = -\frac{mv_x^2}{L} \hat{x}.$$

Note that this is the average force acting on the molecule, induced by Δp_x for every time interval Δt that the molecule undergoes in a round trip for the cylinder. The round trip time itself is $\Delta t = 2L/v_x$, which we have used in the equation. Note also that we let L be the width of the container excluding the outside of the piston.

By Newton's Third Law, the molecule induces an average force on the piston $\langle \mathbf{F} \rangle = -\langle \mathbf{F}_{\text{piston}} \rangle$. If we consider all molecule, we need only to sum over all forces, such that

$$\langle \mathbf{F}_{\text{all}} \rangle = \sum_{k=1}^N \langle -\mathbf{F}_{\text{piston}} \rangle = N \frac{m \langle v_x^2 \rangle}{L}.$$

Here $\langle v_x^2 \rangle$ is the average square velocity. In general, we assume that the velocities are different from each other, inducing an associated average pressure

$$\langle P \rangle = \frac{\langle F_{\text{all}} \rangle}{A} = N \frac{m \langle v_x^2 \rangle}{LA}$$

and, since the volume of the contained occupied by the gas is $V = LA^1$. We can rewrite this equation in the form

$$\langle P \rangle V = Nm \langle v_x^2 \rangle.$$

A direct comparison with the ideal gas equation² (1) shows that

$$Nm \langle v_x^2 \rangle = kNT \implies \frac{m \langle v_x^2 \rangle}{2} = \frac{kT}{2} = \langle U_x \rangle.$$

¹I think this is a 3D example hidden in 2D, where L is the width of the container and A is the area of the piston. That is, the 1D area of the piston is its length.

²Note we also choose form in N , since we are already working with the number of particles, not number density.

In fact, each DOF of the particles has an average energy $\langle U_i \rangle = kT/2$ ($i = x, y, z$), so that the total kinetic energy for a single molecule is

$$\langle U \rangle = \frac{1}{2}m \langle v_x^2 \rangle + \frac{1}{2}m \langle v_y^2 \rangle + \frac{1}{2}m \langle v_z^2 \rangle = \langle U_x \rangle + \langle U_y \rangle + \langle U_z \rangle. \quad (4)$$

We can assume that distribution of velocities is isotropic, and so it follows that

$$\langle U \rangle = \sum_{i=1}^3 \frac{1}{2}kT = \frac{3}{2}kT \quad (5)$$

which implies a root mean square velocity

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3kT}{m}}. \quad (6)$$

For a nitrogen molecule at room temperature, we have the following numerical values: $T = 300$ K and $m_{N_2} = 28$ u, where the atomic mass unit $1 \text{ u} = 1.67 \cdot 10^{-27}$ kg. Computing v_{rms} specifically, we find that $v_{\text{rms}} = 515$ m/s for nitrogen. In fact, we expect the thermal velocities of atoms and simple molecules at room temperature to be in the range of hundreds of meters per second.

1.3 The Equipartition Theorem

The average energy for a single DOF establishes an important link between microscopic and thermodynamic quantities. A similar observation which connects specific heat measurements to thermodynamic quantities by $C \approx 3Rn$.

To be more proper, we define the specific heat of a physical system as

$$C \equiv \frac{Q}{\Delta T} \quad (7)$$

where Q is the heat transferred to an object and ΔT is the change in temperature. The transferred heat is equal to the average change in energy by $Q = \langle \Delta U \rangle$. Then, from the approximation $C \approx 3Rn$, we find that

$$\langle \Delta U \rangle = C \Delta T \approx 3Rn \Delta T = 3R \Delta T \frac{N}{N_A}.$$

Let us assume (incorrectly) that this law can be extrapolated to the point $T = 0$ so that $\Delta T = (T' - T) \rightarrow T'$ where T' is some final time. So, we conclude (by $k = R/N_A$) that

$$\langle \Delta U \rangle = 3k \Delta T N \implies \langle E \rangle = \frac{\langle \Delta U \rangle}{N} = 3kT. \quad (8)$$

(8) shows that the average energy $\langle \Delta U \rangle$ of each atom is given by $3kT$. Notice the similarity of this result to that in §1.2.1, where the average kinetic energy is given by $3kT/2$ for a single molecule. We can use these results to develop a theorem for similar cases.

The Equipartition Theorem. Given some temperature T , the average kinetic energy of some quantity is $kT/2$ for each quadratic degree of freedom.

We can use this theorem to explain the Dulong-Petit law (the statement that $C \approx 3Rn$). For an atom, we can associate a kinetic and potential energy term in each direction. That is,

$$\langle E_i \rangle = \langle K_i \rangle + \langle U_i \rangle = \frac{1}{2}m \langle v_i^2 \rangle + \frac{1}{2}k_s x_i^2, \quad i = x, y, z.$$

Notice that there is a $kT/2$ contribution for *each* quadratic degree of freedom. That is, since $\langle K_i \rangle \propto \langle v_x^2 \rangle$ and $\langle U_i \rangle \propto x_i^2$, we have

$$\langle E \rangle = \sum_{\alpha=1}^3 \langle E_i \rangle = \sum_{\alpha=1}^3 [\langle K_i \rangle + \langle U_i \rangle] = \sum_{\alpha=1}^3 \left[\frac{1}{2}kT + \frac{1}{2}kT \right] = 3kT.$$

Consider now a molecular gas. We have now a contribution of rotational degrees of freedom. For example, around the x -axis, the rotational kinetic energy is given by $I_x \omega_x^2/2$, where I_x is the moment of inertia in x and ω_x is the angular velocity in x . We know that $\langle R_x \rangle \propto \omega_x^2$, (let R_x be the rotational kinetic energy), so that there is an additional $kT/2$ contribution to the molecule's energy.

In contrast, an atom has a degree of freedom number of $f = 3$, which correspond to the three directions in space. This is exactly what we found earlier, with $\langle E \rangle = 3kT$. Even despite the fact that there are clearly more than three electrons and nucleons with their own degrees of freedom, from a classical (and quantum) view, the degrees of freedom with energies $\lesssim kT$ become frozen out.

A more interesting example is a diatomic molecule at room temperature. Here, we have the usual $f_{\text{cart.}} = 3$ from the three Cartesian directions, but now since the diatomic molecule can rotate about two rotationally asymmetric axes, we have the additional terms $\langle R_x \rangle$ and $\langle R_y \rangle$ (or, $\langle R_x \rangle$ and $\langle R_z \rangle$, or similar).

Indeed, we can develop a general formula for the **total thermal energy**, which would be

$$\langle E_{\text{therm.}} \rangle = \frac{1}{2} N f k T \quad (9)$$

where we, of course, assume the presence of no other non-quadratic energies.

There are two very important limitations to the equipartition theorem. In particular:

1. the theorem does not hold well when $T \rightarrow 0$.
2. the theorem does not account for the temperature dependence on the number of degrees of freedom. In fact, the "active" degrees of freedom are those with excitation energies comparable or smaller than the temperature itself.

1.4 Heat and Work

We now state the **first law of thermodynamics**. That is, *the change in energy of a physical system is the sum of the heat transferred to the system and the work done on the system*. Namely,

$$\Delta U = Q + W \quad (10)$$

where Q is the heat transferred as before, and W is the work done on the system. This law is essentially a statement of conservation of energy for the system and its environment.

We can interpret this law as a way to calculate the change in energy of a physical system based on energy transferred from other sources. You might ask: why do we not just calculate the energy change by direct observation? The answer is we can't. It's essentially impossible. We are only able to accurately model physical systems, and thermodynamics handles large systems for which

our models become too convoluted to use. Another issue is the possibility of a microscopic model being hard to understand.

All this is such that in thermodynamics, large systems tend to be in special thermal equilibrium states, described by a small number of thermodynamic variables.

Furthermore, the first law assumes (implicitly) the additivity of energy. We assume that the energies of interactions between sub-systems to be negligible. Is this necessarily true? No. In both classical and quantum physics, the interaction energies between sub-systems can become too significant to ignore.

More rigorously, we may say that the contact area multiplied by the interaction range is negligible relative to the volume of the interactive systems. We may, however, refine this approximation when considering problems between two phases of matter; this is an example of where we might need to introduce energy terms arising from contact surfaces. Think, for example, the coexistence of normal and superconducting regions in a metal.

The assumption of additivity is made for entropy as well. We discuss this later.

Let us now make the distinction between heat and work, both of which are ways to transfer energy to and from a system of interest. Recall that we defined heat as the spontaneous transfer of energy between two objects *due* to a temperature difference. That is, heat flows from higher to lower temperature.

On the contrary, work is energy transferred through any means other than heat. It comes, however, in many forms. For example, **compression work** is the work done by a displacement of the piston that overcomes the pressure of the gas. Indeed, $W_{\text{comp.}} = Fd$. **Electrical work** is the work arising due to a time varying magnetic field to a metal. To illustrate this, consider Fig. 5.

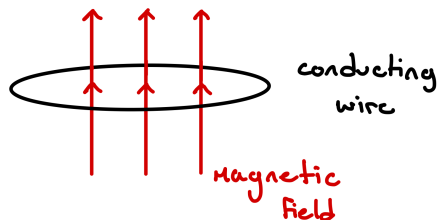


Figure 5: Conducting ring with a homogeneous applied magnetic field.

From Faraday' law, an induced voltage within the ring is given by

$$V = -\frac{d\phi}{dt}$$

where ϕ is the magnetic flux through the ring. Furthermore, we know that the flux is given by

$$\phi = BA$$

where BA is the applied magnetic field and A is the area of the ring. We can thusly express the voltage (assuming A is constant)

$$V = -A\frac{dB}{dt}$$

So then, by Ohm's law we have $V = IR$ and $P = V^2/R$ (P is power, here), so that the work done is P/t . That is,

$$W_{\text{elec.}} = \frac{A^2}{Rt} \left(\frac{dB}{dt} \right)^2.$$

Here, the energy transferred to the electrons is dissipated as a result of electron collisions with atom in the metal. Thus, the temperature of the solid lattice increases (here the metal is the ring). Notice that in this case the energy transfer arises from work. We must be careful to distinguish between the two as they become a source of confusion.

We might ask now if the amount of transferred heat difference since the temperature is different between the systems (the wire loop and the magnetic field generating system). More specifically, does the heat transfer through this process stop at some finite value? The answer is no. Heat is transferred for any temperature so long as the temperature is within a reasonable range.

Often, we can think of work in terms of an agent that interacts with our system of interest. In the compression work case, a mechanism is assumed to control the force on the piston, which in turn induces changes in position. In the electric work case, a magnetic field interacts with spins within a paramagnet; we can exploit this by inducing change in the magnetic field and thereby the current within the coil.

We can also look at work as being associated with changes in parameters of a physical system. When these parameters change, the energy changes, and thereby there is work done on the system. Consider volume, for example, which can be thought as a boundary condition within a physical system. In energy terms, we can say that the potential energy of the boundaries become arbitrarily high, preventing particles from escaping the volume. In quantum mechanics, the change in energy can be associated with changing energy levels due to a varying volume size. In classical mechanics, energy transfer arises from the piston reducing the volume.

Consider now the work due to a magnetic field applied to a system. This is another kind of parameter which we can analyze. Here, we have the energy of a dipole in a magnetic field to be

$$E = -\boldsymbol{\mu} \cdot \mathbf{B}.$$

If $\mathbf{B}$ varies, a larger orientation of the dipoles can be induced, increasing the magnetization. Thusly,

$$\delta R = H \delta \int M dV$$

which is associated with work. In general, the energy of a system depends only on state variables assuming thermal equilibrium. For example, the energy of a gas can be written as a function $U = U(T, N, V)$, where it is implied that the system has reached equilibrium. Although the difference in energy depends only on the initial and final state of the system, the work and heat depends on the path for changes in state variables.

An important type of process in thermal physics are what we call **quasistatic processes**, defined as processes that are so slow that the state is very close to equilibrium for all time. It is complicated to find criteria for a process to be quasistatic. On the other hand, we can find general strategies to deal with non-quasistatic processes.

- Bounds on the work or heat can be derived, and they can be used.
- Work can be calculated based on the energy spent by the agent realizing the work process.

- Heat can be calculated based on missing heat from a physical system in contact with the system of interest; this may be easier if the additional system undergoes a quasistatic process.
- Processes without heat transfer (adiabatic) are physically realizable.

Regardless, the units of energy are defined to be the Joule. That is, $[E] = [J] = [kg][m]^2[s]^{-2}$. Another important unit is the calorie, [cal], which is the energy to increase the temperature of a single gram of water by a single Celsius. In this way, the connection between the Joule and the calorie is the conversion

$$1 \text{ cal} = 4.184 \text{ J.} \quad (11)$$

1.4.1 Mechanical Work

Classical mechanics shows that the work done on an object applying a force, $\mathbf{F}$, with an induced displacement of $\mathbf{d}$ is

$$W = \mathbf{F} \cdot \mathbf{d}. \quad (12)$$

For small displacements, we have the equivalent

$$\delta W = \mathbf{F} \cdot d\mathbf{r}$$

where δW indicates a small, non-differential quantity. Consider again now Fig. 4. The work done on the gas by the piston is given by

$$W = F^{(e)} \Delta x$$

where $F^{(e)}$ denotes the external force. When changes are slow, the external force overcomes the force due to the pressure acting on the opposite side of the piston only slightly. In fact, we have

$$F^{(e)} \approx PA \quad (13)$$

which assumes that P is well-defined and is the same at the piston surface and in the bulk of the gas. On the other hand, we require $F^{(e)} > PA$ for the piston to start moving. Let us assume that the change on the piston is slow. Then, we have that

$$W \approx PA\Delta x = -P\Delta V$$

where we have used $A\Delta x = \Delta V$. Note the sign: positive works corresponds to gas compression. Further, we might be interested in what the expected order of magnitude is. Suppose we have 1 atm $\approx 10^5$ Pa, $\Delta V = -1$ L. Then, we have $W = -(10^5 \text{ Pa})(-0.001 \text{ m}^3) \approx 100 \text{ J}$.

Consider now a process carried out at constant pressure shown in Fig. 6. Recall that the work done is given by $W = -P\Delta V$. Here, the work would be represented as the area under the curve in both (a) and (b). Notice, also, that the sign of the work is governed by the motion of the curve. In (a) rightward motion corresponds to negative work, and hence the volume from $V_i \rightarrow V_f$ increases. In (b), leftward motion corresponds to positive work, and hence the volume from $V_i \rightarrow V_f$ decreases.

We can, for a more general process, assume that the pressure is non-constant as represented in Fig. 7. As a rough approximation, we may say that the work is given by the sum

$$W \approx - \sum_{j=1}^n P_j \Delta V_j.$$

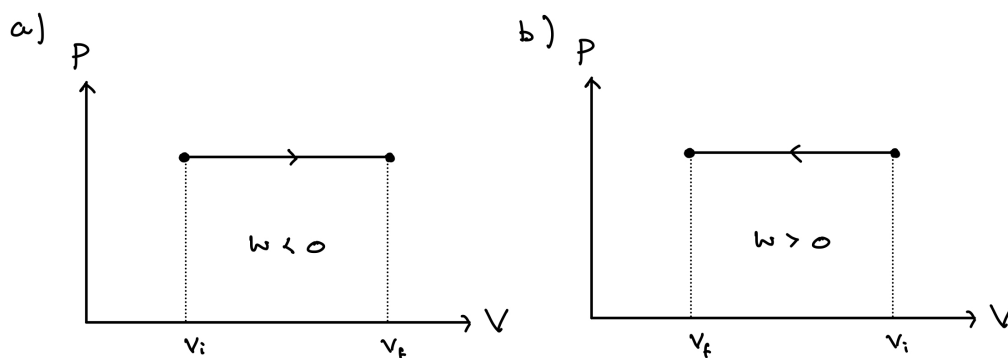


Figure 6: Constant pressure compression as a function of volume. The work done is represented by the area under the curve.

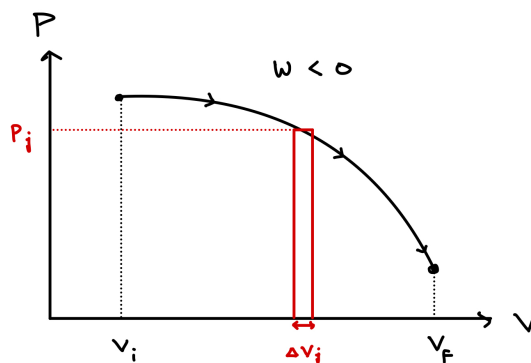


Figure 7: Non-constant pressure as a function of volume for a compressed gas.

And so naturally, when we consider the limiting case $\Delta V_j \rightarrow 0$, we arrive at the integral formulation

$$W = - \int_V P dV \quad (14)$$

1.5 Mechanical Work for an Ideal Gas

Let us now analyze the ideal gas for different types of processes.

1.5.1 Isothermal Compression

We define **isothermal compression** to be the compression process at some constant temperature. *Iso-* meaning "single". Fig. 8 illustrates how pressure depends on volume during this process.

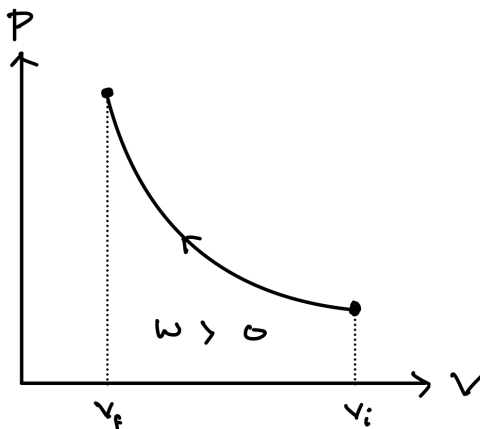


Figure 8: An isothermal process represented as a PV (pressure-volume) diagram.

The work is given by the integral formulation

$$W = - \int_{V_i}^{V_f} P dV$$

and, of course, note that we expect the work done to be a positive quantity. If the process is *quasistatic*, the pressure of the agent inducing the work is the same as the internal pressure of the gas, allowing us to model this situation with the ideal gas law. That is, we have that $P = kNT/V$ implies

$$W = - \int_{V_i}^{V_f} \frac{kNT}{V} dV = -kNT \int_{V_i}^{V_f} \frac{dV}{V} = kNT \ln\left(\frac{V_i}{V_f}\right) > 0.$$

This agrees with what we expect. In fact, $V_i > V_f \implies \ln(V_i/V_f) > 0$ so that positive work is done. We are interested in how this affects the change in energy and the transfer of heat for the process. Considering first the energy, we know that the energy is dependent only on temperature, and so since $\Delta T = \Delta U = 0$, the first law of thermodynamics shows that

$$Q = \Delta U - W = -W = -kNT \ln\left(\frac{V_i}{V_f}\right) = kNT \ln\left(\frac{V_f}{V_i}\right).$$

Here is a neat observation: for a quasistatic isothermal process, we have that $Q = -W$, and so the heat transferred will always be a result of positive work.

1.5.2 Adiabatic Compression

We define **adiabatic compression** as a process for which there is no heat transferred. Indeed, $Q = 0$. The first law of thermodynamics shows that in this case,

$$\Delta U - W = 0$$

and for small cases,

$$dU - \delta W = 0.$$

It is for this reason that we define a "small change" (and not differential) by the symbol δ , and not the derivative. In fact, the differential change in energy, dU , depends only on the change in state, while the work done, δW , depends only on the path for the process.

If we recall the general expression for total thermal energy by (9), we find that

$$U = \frac{1}{2}NfkT \implies dU = \frac{1}{2}Nfk dT$$

and we can combine this with the small work done

$$\delta W = -P dV$$

to arrive at the differential form of the first law of thermodynamics,

$$\frac{1}{2}Nfk dT + P dV = 0. \quad (15)$$

It is worth noting that the system state can be described in terms of T and V . (15) also allows us to understand the slope of the $T(V)$ dependence; in fact, this slope is enough to determine the governing $T(V)$ equation. To do this, we must express the coefficients of the differential form in terms of only T and V . The ideal gas law does just this: $P = kNT/V = P(T, V)$ since N and k are constants, and so we can solve (15) to find

$$\begin{aligned} \frac{1}{2}Nfk dT + \frac{NkT}{V} dV &= 0 \\ \frac{1}{2}f \int_{T_i}^{T_f} \frac{dT}{T} &= - \int_{V_i}^{V_f} \frac{dV}{V} \\ \frac{f}{2} \ln\left(\frac{T_f}{T_i}\right) &= - \ln\left(\frac{V_f}{V_i}\right) \\ \left(\frac{T_f}{T_i}\right)^{\frac{f}{2}} &= \left(\frac{V_i}{V_f}\right). \end{aligned}$$

Notice that this result is constant. In fact, we may wish to express this in the form

$$V_f T_f^{\frac{f}{2}} = V_i T_i^{\frac{f}{2}} \implies VT^{\frac{f}{2}} = C \quad (16)$$

for some constant C . To use this to calculate work, we can re-express this in terms of P and V by the ideal gas law once more,

$$V \left(\frac{PV}{kN} \right)^{\frac{f}{2}} = C \implies P^{\frac{f}{2}} V^{\frac{f}{2}+1} = kNC$$

Taking the inverse power $2/f$, we thus have

$$PV^{\frac{2+f}{2} \frac{2}{f}} = D$$

for the new constant $D = (kNC)^{2/f}$. This leads us to introduce the **adiabatic exponent**,

$$PV^\gamma = D, \quad \gamma = \frac{2+f}{f}. \quad (17)$$

Now consider the work and energy for an adiabatic process at an initial pressure, P_i , and initial volume, V_i . We have,

$$W = - \int_{V_i}^{V_f} P dV = - \int_{V_i}^{V_f} \frac{P_i V_i^\gamma}{V^\gamma} dV$$

where we have used the relation $P_i V_i^\gamma = P V^\gamma$. Solving this explicitly, we have

$$W = - \frac{P_i V_i^\gamma}{1-\gamma} (V_f^{1-\gamma} - V_i^{1-\gamma}) = \frac{P_i V_i}{\gamma-1} \left[\left(\frac{V_i}{V_f} \right)^{\gamma-1} - 1 \right] = \frac{1}{2} f P_i V_i \left[\left(\frac{V_i}{V_f} \right)^{\gamma-1} - 1 \right].$$

Using $P_i = NkT_i \implies P_i V_i = NkT_i V_i$, we thus have

$$W = \frac{1}{2} f NkT_i \left[\frac{T_f}{T_i} - 1 \right] = \frac{1}{2} f NkT_f - \frac{1}{2} f NkT_i \quad (18)$$

where we have used the relation $\gamma - 1 = 2/f$. Notice that this result implies that $W = \Delta U$ since $Q = \Delta U - W$. That is, $Q = 0$.

1.5.3 Mechanical Work and Energy for Different Processes

The above two processes are only two of the most common processes in thermodynamics. We, however, have other processes such as **isochoric** ($V = \text{constant}$) and **isobaric** ($P = \text{constant}$) processes.

Isochoric processes show that the mechanic work is zero, since it follows that $dV = 0$. On the other hand, isobaric processes have work given by $W = P(V_f - V_i) = P\Delta V$, where P is constant. Note that this result is independent of the type of system used, and is not restricted to only an ideal gas.

Process	Characteristic
Isothermal	$T = \text{constant}$
Isobaric	$p = \text{constant}$
Isochoric	$V = \text{constant}$
Adiabatic	$Q = 0$

Table 2: Common types of processes in thermodynamics.

For an arbitrary quasistatic process, the steps for calculating work, heat, and energy change can be done in a streamlined manner:

- (1) Establish a $P(V)$ dependence for the process and use it to calculate work based on.

$$W = - \int_V p(V) dV$$

- (2) Use an expression for the energy as a function of state variables to calculate the change in energy; that is,

$$\Delta U = U_f - U_i.$$

(3) By the first law of thermodynamics, calculate the heat by

$$Q = \Delta U - W.$$

The only exception to these steps is the adiabatic process.

We make, as a final note, that a non-quasistatic process differs significantly. In fact, we cannot calculate the work by the integral formulation since the pressure by the external agent may differ from the internal pressure of the system; and, the internal pressure does not depend on the volume or other parameters based on a simple equation of state under equilibrium.

1.6 Heat Capacity

Recall the definition for the **heat capacity** of a physical system,

$$C \equiv \frac{Q}{\Delta T}$$

where ΔT is the change in temperature resulting from an object receiving a quantity of heat, Q . The heat capacity is dependent on the state variables in initial conditions as well as the type of process occurring during heat transfer. However, it should be noted that (7) is an *empirical* definition, under the observation that

$$Q \sim \Delta T.$$

That is, the heat is on the order of magnitude of the temperature difference. In infinitesimal quantities, we can redefine the heat capacity to be in the form

$$C \equiv \lim_{\Delta T \rightarrow 0} \frac{Q}{\Delta T} = \frac{\delta Q}{dT}. \quad (19)$$

Notice we have distinguished between a differential and a "small quantity" again with δ . In general, we assume that the heat capacity of a system is given by (19), unless otherwise specified. Relatedly, we can define the **specific heat capacity** to be

$$c \equiv \frac{C}{m} \quad (20)$$

where C is the heat capacity and m is the mass of the system. This is also an empirical definition, motivated by the observation that heat transferred to an object inducing a temperature change is proportional to the size and (equivalently) the mass of the object. Fundamentally, we can connect this to the additive property of energy which we have assumed previously.

Heat transferred to an object is also path dependent aside from being temperature induced. In fact, by invoking the first law of thermodynamics, we find that

$$C \equiv \frac{Q}{\Delta T} = \frac{\Delta U - W}{\Delta T} = \frac{\Delta U}{\Delta T} - \frac{W}{\Delta T}. \quad (21)$$

Here, (21) shows that the heat capacity is dependent on both a path-dependent quantity (work) and a state-dependent quantity (energy). We should further note that the energy quantity $\Delta U/\Delta T$ is not only dependent on the initial and final temperatures of the system, as the state of the system itself can be governed by other parameters.

1.6.1 Heat Capacity for Specific Processes

Isochoric Processes. Let us consider the case where volume is constant. It follows that the work on the system is zero, $W = 0$, and so we have a heat capacity

$$C = \frac{\Delta U}{\Delta T}$$

and in the limit that we take the infinitesimal, we have the more accurate definition

$$C = \left(\frac{\partial U}{\partial T} \right)_V = \lim_{\Delta T \rightarrow 0} \frac{U(T + \Delta T, V) - U(T, V)}{\Delta T}. \quad (22)$$

Isobaric Processes. Let us consider now the case where pressure is constant. Here, a change in volume on the system would induce a work given by $W = -P\Delta V$, and so we result at a heat capacity

$$C = \left. \frac{Q}{\Delta T} \right|_P = \left. \frac{\Delta U}{\Delta T} \right|_P + P \frac{\Delta V}{\Delta T}.$$

Again, we take the limit as $\Delta T \rightarrow 0$, and so we arrive at the partial differential form

$$C = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P. \quad (23)$$

There are a few remarks we should make about this form. The partial derivative, $(\partial V / \partial T)_P$ is negligible for solids and liquids, but rather important for gases. On the other hand, we should note that the first term is, in general,

$$\left(\frac{\partial U}{\partial T} \right)_P \neq \left(\frac{\partial U}{\partial T} \right)_V.$$

That is, $(\partial U / \partial T)_P$ is not equivalent to C_V .

1.6.2 Specific Heat of Ideal Gases

We can calculate the specific heat of an ideal gas by using the Equipartition Theorem and the ideal gas law. In fact, the general formula for total thermal energy (9) shows that

$$\frac{\partial \langle E_{\text{therm.}} \rangle}{\partial T} = \frac{\partial U}{\partial T} = \frac{\partial}{\partial T} \left(\frac{1}{2} N f k T \right) = \frac{1}{2} f N k.$$

Notice that this does not depend on the volume or pressure of the gas itself. It is a unique result to *any* ideal gas, and so we may obtain from the ideal gas law that

$$\left(\frac{\partial V}{\partial T} \right)_P = \frac{\partial}{\partial T} \left(\frac{N k T}{P} \right) = \frac{N k}{P}.$$

Of course, it follows that the constant volume heat capacity is

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{\partial U}{\partial T} = \frac{1}{2} f N k \quad (24)$$

and for constant pressure heat capacity,

$$C_P = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P = \frac{1}{2} f N k + P \frac{N k}{P} = N k \left(\frac{f}{2} + 1 \right). \quad (25)$$

1.6.3 Latent Heat

In special cases, we may have processes where heat transfer does not induce temperature change. These correspond to **phase transition** of our object. The heat required to cause a complete transition between two phases is called **latent heat**, and we denote this by L . In this way, we define the **specific latent** to be

$$l \equiv \frac{L}{m} \quad (26)$$

which is of extremely similar form as (20). Generally, we will know L to be given as it is an empirical quantity.

1.6.4 Measurements of Specific Heat

Typically, we measure the specific heat of a system by the use of calorimeters. A typical calorimeter uses the fact that an arbitrary system of interest is brought into thermal contact with a holder containing a resistive element and a thermometer. Applied electrical power to the system dissipates as heat in the resistive element, and allows us to measure (infer) the heat capacity by first measuring the temperature using the thermometer.

We can measure latent heat with a calorimeter as well. However, it is important to be clear about where the phase transition occurs in our system. We can do this by observing the temperature of the system, as the coexistence of two phases can be identified based on the period of constant temperature despite the addition of heat. Alternatively, we may opt to use optical observations, viscosity, or electrical resistance.

We typically measure specific heat during an isobaric process, since it is impractical to constrain the volume of our object. For water, specifically, at an ambient pressure of 1 atm, the values of specific and latent heat are

$$\begin{aligned} C_l &= 4.2 \text{ Jg}^{-1} = 1 \text{ cal g}^{-1}. \\ C_{s \rightarrow l} &= 333 \text{ Jg}^{-1} = 80 \text{ cal g}^{-1} \\ C_{l \rightarrow g} &= 2260 \text{ Jg}^{-1} \end{aligned}$$

where we denote l to be the liquid phase, $s \rightarrow l$ to be the solid-liquid transition (melting), and $l \rightarrow g$ to be the gas transition.

1.7 Enthalpy

Isobaric processes are extremely important in many controlled experiments and, in general, show up in many physical processes. In discussing these processes, we must introduce the quantity called **enthalpy**. We define enthalpy as the following

$$H \equiv U + PV \quad (27)$$

so that at constant pressure, we have the change in enthalpy

$$\Delta H = \Delta U + P\Delta V. \quad (28)$$

We will cite the second law of thermodynamics (which we will discuss later) to show that the energy ΔU is composed of the energy of the entire system (including the environment) and any external

work. That is,

$$\Delta U = Q + W + W_{\text{ext.}}$$

We must be clear in making the distinction between *mechanical work*, W , and *non-mechanical* (external) work, W_{other} . The mechanical work we define as $W = -P\Delta V$, so we can, by the first law of thermodynamics, write the enthalpy change as

$$\Delta H = Q + W_{\text{ext.}} \quad (29)$$

Fig. 9 shows a graphical representation of how enthalpy plays a role in isobaric systems.

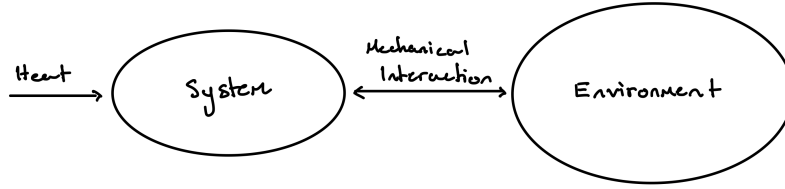


Figure 9: System coupled to its environment by mechanical interaction due to initial transferred heat.

Indeed, the system we are interested in and its environment create a combined, coupled system. Notice that while heat is transferred by coupling alone, it induces a total energy change between the system *and* the environment. That is,

$$Q = \Delta(U_S + U_E) = \Delta U_S + \Delta U_E \quad (30)$$

where U_S is the energy of the system and U_E is the energy of the environment. We may argue that the environment is generally much larger than the system, so the finite volume change in the environment is negligible in induced pressure. That is, $\Delta V_E \ll \Delta V_S \implies P_E = \text{const.}$ Further, since volume of the entire system must remain constant (the subsystems may differ), we also have

$$\Delta V_E + \Delta V_S = 0.$$

Assuming a quasistatic process, we thus have pressure equality $P_S = P_E$, and so we can write the heat transferred as

$$Q = \Delta U_S + \Delta U_E = \Delta U_S - P_E \Delta V_E = \Delta U_S + P_S \Delta V_S = \Delta H_T \quad (31)$$

where we denote H_T to be the enthalpy of the entire system. Notice how this contrasts with the case of an isolated system. There, applied heat would result in an equal change in the object system's energy, U_S . In a coupled environment, we instead have a change in pressure, which induces work on the environment due to varying boundary conditions.

There are thus two effects: (1) increase in energy to the system. (2) Increase in energy of the environment, both of which are suggested by (31) under the assumption of a quasistatic process. We can further generalize this result to the presence of non-mechanical work, $W_{\text{ext.}}$. Indeed, we need only to introduce the energy energy as

$$\Delta H_T = Q + W_{\text{ext.}} \quad (32)$$

1.7.1 Enthalpy and Specific Heat in Isobaric Processes

Recall that for an isobaric process, the heat capacity is given in the form

$$C = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P.$$

This derivation assumes that all thermodynamic variables are expressed in terms of temperature and pressure. In terms of enthalpy, we have a similar form,

$$C_P = \left(\frac{\partial H}{\partial T} \right)_P = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P. \quad (33)$$

Notice that this implies that the specific heat capacity is related to the increase in enthalpy. In fact, heat is spent to both increase the energy of our system at a fixed pressure.

1.7.2 Analysis of Chemical Reactions by Enthalpy

Let us suppose we are interested in the formation of water, governed by the reaction



If we are given that (by known quantities) the change in enthalpy for the reaction is $\Delta H = -286$ kJ per mole, we have two possible interpretations:

- (1) enthalpy is such that $\Delta H = Q < 0$. That is, during the reaction, heat is extracted from the system (formed by the reactants and their products), while the environment is a physical system that maintains constant pressure.
- (2) If the reaction takes place in a thermally insulated container, we have $Q = 0$, which seemingly contradicts the observation $\Delta H < 0$.

We can solve the issue in the second interpretation by considering a small step in the reaction, where the quantity of the reaction product increases from x ($0 < x < 1$) moles to some $x + \delta x$ ($\delta x \ll 1$) moles. The heat, $-Q$, that would normally be evacuated here becomes dumped into the system of reactants and products, and we can represent this by

$$\delta x \Delta H(T) = -c_\nu(x, T) \delta T = Q. \quad (35)$$

We stress that δx here is a small change in the converted reactant fraction measured in moles, and ΔH is the change in enthalpy for a complete change of a mole of reactants to products. That is, the change in enthalpy per mole depends on the temperature.

Furthermore, at each step, we have a change in temperature induced by the heat given by the product of the specific heat $c_\nu(x, T)$ for a mix of $1 - x$ moles of reactants and x moles of reaction products, also dependent on temperature. That is, since the heat is negative and the temperature is positive, the reaction is **exothermic**.

1.8 Elements of Kinetic Theory

While we will be focusing on systems during quasistatic processes, systems that operate in non-equilibrium states are also extremely important. Every system in a non-equilibrium state has a natural tendency to restore equilibrium, be it by redistribution of energy, redistribution of particles, etc.

Away from equilibrium, we can describe the dynamics of systems by **transport phenomena**. Table 3 shows a quick summary of the types of transport phenomena we may in other systems.

Phenomenon	Equivalent "force"	Transported Quantity
Diffusion	Chemical potential	Number of particles/mass
Electric current	Electrochemical potential	Electric charge
Heat transport	Temperature	Energy
Viscosity	Momentum gradients	Momentum density
Spin transport	Spin chemical potential bias	Spin

Table 3: Examples of transport phenomena in other systems.

1.8.1 Heat Conduction

Fourier's Law. We begin with developing **Fourier's Law of Heat Conduction**. Consider a conductive process, where heat transfer processes at a microscopic level. We can understand conduction in gases and liquids as the result of intermolecular collisions. In solids, heat transfer may occur by a propagation of energy in the form of sound waves or collisions between free electrons (for metals and semiconductors).

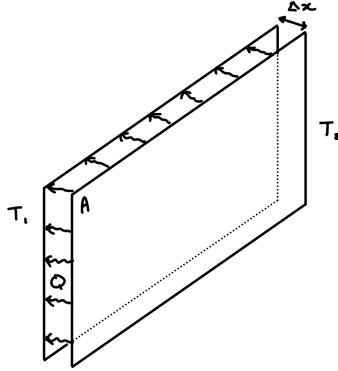


Figure 10: Two planes of equal area, A , separated by a small distance Δx , under the process of heat conduction.

In Fig. 10, we seek to analyze heat transfer across a plane of area, A , under some finite temperature change, ΔT , and a varied distance, Δx . Empirically, first and foremost, the transferred heat is suggested to be given by a form

$$Q \propto \Delta t \Delta T \frac{A}{\Delta x} \quad (36)$$

where Δt is the time interval over which we measure the heat. Let us analyze how these parameters affect our system.

- Δt : We observe that changes in temperature of a region that receives heat evolves linearly in time.
- A : We observe that heat transfer is proportional to the size of our system. Indeed, it is proportional to the contact area.
- ΔT : We already know that heat transfer must depend on some finite temperature difference. Of course, this suggests a differential equation for the temperature difference given by $d(\Delta T)/dt \sim \Delta T$ which has an exponential solution.
- $1/\Delta x$: We observe that scaling the temperature difference and the distance between the planes by the same factor does not change the transferred heat. That is, we should expect $\Delta T/\Delta x \sim 1$.

Taking (36) to its differential form, we need only to bring $\Delta x \rightarrow 0$ in a limit. Then, we have

$$\frac{Q}{\Delta t} = -k_t A \frac{dT}{dx} \quad (37)$$

where k_t is an introduced constant called the **heat conductivity**, specific to any substance generally dependent on temperature and some other variable. (37) is the Fourier's Law of Heat Conduction which we sought to find. Notice, we have also introduced a negative sign. To understand this, consider Fig. 11.

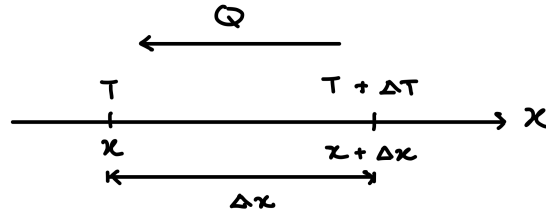


Figure 11: Flow of heat for Fourier's heat conduction law. Positive gradient corresponds to negative heat flow (in the negative x -axis direction).

Here, a positive gradient in the temperature (along x) implies a transfer of heat in the negative direction of the x -axis.

Heat Conductivity of an Ideal Gas. Consider now an ideal gas. Here, the density of the molecules within the gas is low, and intermolecular distances are much larger than molecular sizes. We are not interested in the collisions (as they are unnecessary for describing the gas state), however we must incorporate them when modelling heat conduction.

Assuming a hard sphere model where each molecule has a radius, r , we can infer that if their centres are a distance shorter than $2r$, a collision will occur. In particular, we can consider a simplified picture where a single molecule moves and the other remain at rest (Fig. 12). Here, we denote only the *centre* of the molecule with a dot.

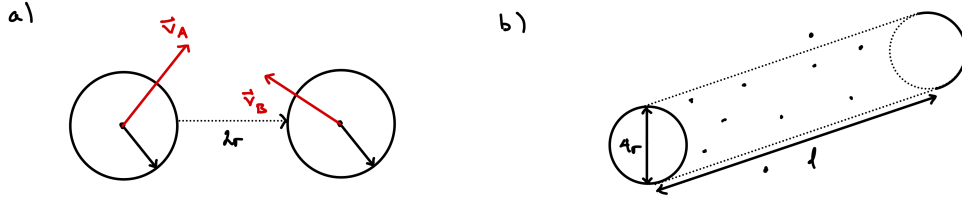


Figure 12: (a) Hard sphere model of two molecules on a possible collision course. (b) Collisional path of a molecule of radius, r , sweeping out a cylinder.

Notice that a typical distance travelled, say l , is such that we have

$$4l\pi r^2 = \frac{V}{N}$$

where V is the volume of the cylinder and N is the number of molecules. Thus, we can conclude that

$$l = \frac{V}{4\pi N r^2} \quad (38)$$

is the mean free path of the sweeping molecule. All molecules within a gas, however, move with random velocities. We can modify our simplified model by considering the motion of the molecule of interest, say A , in a new reference frame where molecule B is at rest. In this way, we can consider different "sweeping volumes", each with sweeping volume corresponding to A and another molecule within the gas.

Regardless, it is known that a typical molecule size is given by $r = 1.5 \text{ \AA}$. Assuming a room temperature gas ($T = 298 \text{ K}$), and an ambient pressure of $P = 1 \text{ atm}$, we have by the ideal gas law

$$\frac{V}{N} = \frac{kT}{P} = 4 \cdot 10^{-26} \text{ m}^3$$

and so our mean free path is $l \approx 150 \text{ nm}$. We can compute the average time between collision, which is

$$\langle \Delta t \rangle = \frac{l}{\langle v \rangle} \approx \frac{l}{v_{\text{rms}}} \approx 3 \cdot 10^{-10} \text{ s}.$$

For heat conductivity, we can consider a gradient in, say, the x -direction as in Fig. 13. Here, half of the molecules in ($v_x > 0$) in the region $x < x_0$ cross the surface if they are within the mean free path, l , of the boundary. Further, the volume of the molecules (as a whole) is lA with an energy of $U_1/2$, where U_1 is the total energy in said volume. We attach a $1/2$ factor to account for the fact that only molecules with $v_x > 0$ contribute.

Similarly, we have an energy term U_2 transported from left to right, with $U_2/2$ being the total energy in a volume of lA at the right of $x = x_0$. From the first law of thermodynamics (we have $W = 0$), it follows that

$$Q = \frac{1}{2}(U_1 - U_2) = \frac{1}{2}c_V(T_1 - T_2)lA = -\frac{1}{2}c_V\left(\frac{T_2 - T_1}{\Delta x}\right)Al^2$$

where $\Delta x = l$ is the distance between middle points of the relevant volumes. Over a time interval Δt , we determine the heat transfer by molecules with an average velocity $\langle v \rangle$ over a distance l . Thusly,

$$\frac{Q}{\Delta t} = -\frac{1}{2}c_V l \frac{dT}{dx} Al^2. \quad (39)$$

Naturally, we obtain the thermal conductivity coefficient by comparing (39) to (37), yielding

$$k_t = \frac{1}{2} c_V l \langle v \rangle. \quad (40)$$

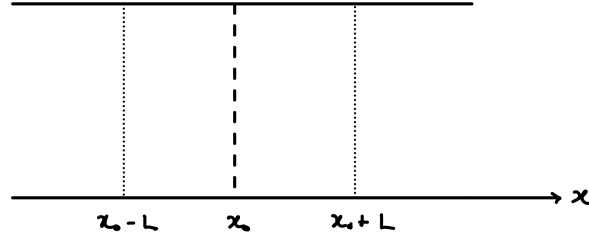


Figure 13: Heat conductivity model based on energy transfer through a surface at $x = x_0$. Here, $L = l$ to indicate a section of the mean free path.

For an ideal gas, we have that

$$c_V = \frac{C_V}{V} = \frac{fkN}{2V} = \frac{fP}{2T}. \quad (41)$$

That is, we find the thermal conductivity coefficient to be

$$k_t = \frac{1}{2} \frac{fP}{2T} \frac{V}{4\pi Nr^2} \langle v \rangle \propto \frac{1}{2} fk \langle v \rangle \propto \sqrt{T}. \quad (42)$$

This result is in agreement with the experimental observations for many gases.

1.8.2 Viscosity

We next discuss *viscosity*, which is a phenomenon that corresponds to the transfer of momentum. Consider the situation shown in Fig. 14. Here, a fluid is trapped between two parallel plates, where one is kept fixed and another moves at velocity $\mathbf{v}_m$. After some finite time, the fluid begins moving so that the velocity is zero next to the bottom plate and $\mathbf{v}_m$ next to the top plate.

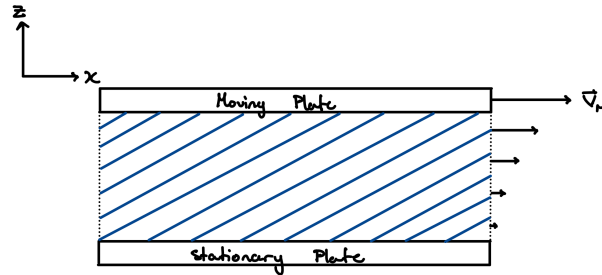


Figure 14: A fluid contained between two parallel plates, one at rest and one moving at velocity $\mathbf{v}_m$. In steady state, we establish a constant velocity gradient.

In this way, we establish a constant velocity gradient in steady state. To analyze this, consider a molecule at a position $z = z_0$ as in Fig. 15.

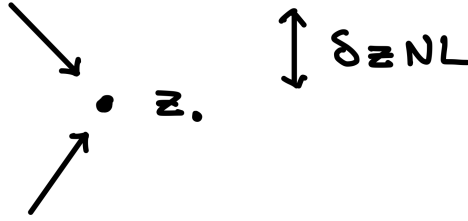


Figure 15: Molecule situated at $z = z_0$ between two plates of the system shown in Fig. 14.

Molecule in a layer above with a thickness δz transfer a momentum

$$P_{\text{up}} = \frac{1}{2} m \left. \frac{dv_x}{dz} \right|_{z_0 + \delta z/2}$$

where the $1/2$ factor accounts for the linear increase in relative velocity of the molecules with height z , thus leading to an average that corresponds to half the relative velocity at height $z_0 + \delta z$. We similar have a momentum transfer in the layer below with

$$P_{\text{down}} = -\frac{1}{2} m \left. \frac{dv_x}{dz} \right|_{z_0 + \delta z/2}.$$

In steady state, each molecule experiences collisions where, on average, positive and negative momentum are transferred. As a result, we reach a balance in the momentum transfer and the velocity gradient is constant.

Consider now the molecules just below the top plate. In the x -direction, a momentum is transferred such that

$$\Delta \mathbf{P}_{\text{molecules} \rightarrow \text{plate}} = -\frac{1}{2} m \left(\frac{dv_x}{dz} \right) \frac{N}{V} l A \hat{x}.$$

As before, we assume that the molecules occupy a volume Al , where A is the area of the plates and l is the mean free path. Thus, the number of molecules that contribute to momentum transfer is that of the density N/V , where N is the number of particles and V is the volume of the fluid. We know the average momentum transfer already, and so we can argue that the transfer occurs over an interval $\Delta t = l / \langle v \rangle$, averaged over the mean free path.

We expect a force required to drag the plate, which we find is

$$\mathbf{F} = -\frac{\Delta \mathbf{P}_{\text{molecules} \rightarrow \text{plate}}}{\Delta t} = \frac{1}{2} \frac{m}{\Delta t} \left(\frac{dv_x}{dz} \right) \frac{N}{V} l A \hat{x}.$$

Simplifying further, we introduce the **coefficient of viscosity**, η , such that

$$\frac{F}{A} = \eta \frac{dv_x}{dz}, \quad \eta = \frac{1}{2} m l \frac{N}{V} \langle v \rangle. \quad (43)$$

We note that $l \propto V/N$ and $\langle v \rangle \propto \sqrt{T}$, so we expect that

$$\eta \propto \sqrt{T}.$$

Thus, we conclude that for a gas the viscosity increases as the temperature increases. However, this does not hold for liquids, where the viscosity increases as temperature decreases, which is a direct result of the fact that collision effectiveness is temperature-dependent.

1.8.3 Diffusion

Of course, we are interested in *diffusion*, which is the transport of number of particles or mass. Consider the case of a gas where the density varies along the x -direction, as in Fig. 16.

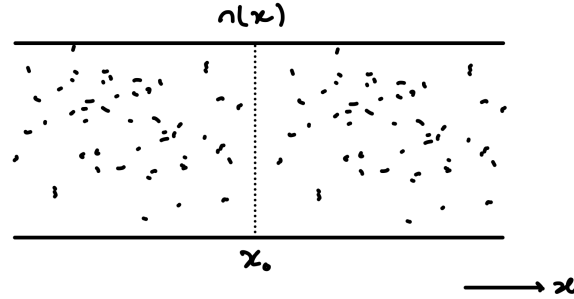


Figure 16: Gas with some variable density $n(x)$.

Defining the particle flux in the x -direction to be $J_x = \Delta N / \Delta t A$, where

$$\Delta N = \# \text{ of particles crossing } x_0 \text{ to the right} - \# \text{ of particles crossing } x_0 \text{ to the left} \quad (44)$$

is the difference between the number of molecules crossing in the two directions, A is the total cross-sectional area of the gas, and Δt is a characteristic time. Similar arguments as in the discussion of thermal conduction, we can write

$$\Delta N = \frac{1}{2} l A \left[n \left(x_0 - \frac{l}{2} \right) - n \left(x_0 + \frac{l}{2} \right) \right],$$

where we considered that molecules within a distance l from x_0 cross the surface, and we let the density of particles, $n(x)$, to be position-dependent. We can make an approximation such that

$$n \left(x_0 - \frac{l}{2} \right) - n \left(x_0 + \frac{l}{2} \right) = -\frac{dn}{dx} l.$$

Of course, the characteristic time is $\Delta t = l / \langle v \rangle$, and so we have that the particle flux is

$$J_x = -\frac{1}{2} l \langle v \rangle \frac{dn}{dx}.$$

We will hide some of the variables within the **diffusion coefficient**, so that

$$J_x = -D \frac{dn}{dx}, \quad D = \frac{1}{2} l \langle v \rangle. \quad (45)$$

In the most general case, (45) becomes **Fick's law**, which is of the form

$$\mathbf{J} = -D \vec{\nabla} n. \quad (46)$$

2 Entropy

Until now, our discussions have been only on the common state variables of our systems of interest, with some elements of kinetic theory. In this section, we begin the discussion of **entropy**, which is directly tied to the number of microstates of our system of interest. Despite this, however, entropy is still well-established in the perspective of macroscopic thermal physics. It has many applications in modern research, such as quantum information, condensed matter, and even cosmology.

In the macroscopic picture, we can look at entropy as a new state variable, S . Microscopic models differ slightly in the discussion of, say, the Einstein solid, or the ideal gas. However, for our scenario, we will focus on the ideal gas as it is the "nicest" for our analysis thus far.

We say that entropy implies **irreversibility**, and we can compactly write the relation

$$S(U, V, N) \longleftrightarrow U(S, V, N)$$

which is an example of a thermodynamic identity. A way of looking at this is that the entropy is a formation into the energy of a system.

Macroscopic and Microscopic States. Macroscopic systems are useful as they highlight the strength of thermal concepts. Observations (which are generally easier to make) in a macroscopic view are directly tied to microscopic models, and so they are more useful in applications. Two important parameters to keep in mind, however, are **microstates** and **macrostates**. In short, we say that the macrostate is a global parameter which links itself to total energy and total volume; further, a single macrostate is linked to *many* microstates.

Consider again the ideal gas. We previously found, from the Equipartition Theorem, that the energy of a molecule is given by $U = f k N T / 2$ *on average*. However, it is possible for a single molecule to have an energy $U' > U$, so there are *many ways* to divide energy.

2.1 The Einstein Solid

Let us consider a system that is slightly more complicated, but well-informed for us to study the physics of thermodynamics. A typical solid is defined as a *set* of atoms (or molecules) with pairwise interactions. We can represent the energy of the solid as the sum of the energies of the atoms,

$$U_p = \sum_{\langle i, j \rangle} U(\mathbf{r}_i, \mathbf{r}_j), \quad U(\mathbf{r}_i, \mathbf{r}_j) = \frac{1}{2} k_{ij} |\mathbf{r}_i - \mathbf{r}_j - \mathbf{r}_{ij,0}|^2 \quad (47)$$

where $\mathbf{r}_i$, $\mathbf{r}_j$, and $\mathbf{r}_{ij,0}$ are positions respectively. Note that $\mathbf{r}_{ij,0}$ is the equilibrium position of a particular particle pair. Note that the define $\langle i, j \rangle$ to be the nearest neighbour interactions only. If we consider a single atom in this solid, we can model it as a oscillating "ball", with a potential energy

$$U_{p,i} = \frac{1}{2} k |\mathbf{r}_i - \mathbf{r}_{i,0}|^2 \implies U_p = \sum_{i=1}^N \frac{1}{2} k |\mathbf{r}_i - \mathbf{r}_{i,0}|^2.$$

which we can naturally extend to a sum over N atoms. For each harmonic oscillator (atom), we thus have that its total energy is its own Hamiltonian,

$$\mathcal{H} = \frac{1}{2} k |\mathbf{r}|^2 + \frac{|\mathbf{p}|^2}{2m}.$$

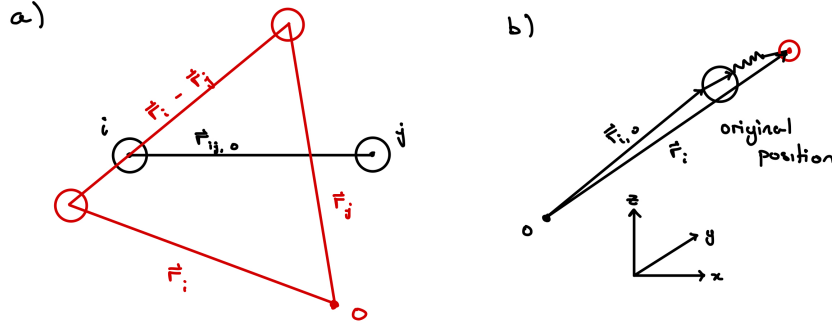


Figure 17: (a) Simple solid with the $\mathbf{r}$ vectors defined. (b) The Einstein model of a single atom.

And, in extension to the Cartesian directions, we have the relations

$$\mathcal{H} = \mathcal{H}_x + \mathcal{H}_y + \mathcal{H}_z, \quad \mathcal{H}_i = \frac{1}{2}kx_i^2 + \frac{p_i^2}{2m}$$

for $i = x, y, z$. We can borrow from quantum mechanics so that for each discrete energy level,

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega_0, \quad \omega_0 = \sqrt{\frac{k}{m}}. \quad (48)$$

for $n \in \mathbb{Z}^+$. So, for M atoms in our system, we have $N = 3M$ harmonic oscillators.

We must discuss the multiplicity of the Einstein solid we are consider, and to do this we will note that the energy of the solid must be $q\hbar\omega_0$ for $q \geq 0$ and $q \in \mathbb{Z}^+$. Note that we offset the energy of the particle to start at $E_0 = 0$.

2.1.1 Multiplicity of the Einstein Solid

Let us first consider a set of N harmonic oscillators (HOs) with a total energy of $q\hbar\omega_0$. We define the number of microstates at the number of ways in which energy can be distributed over the HOs. For example, if we consider 3 HOs (so that $N = 3$), the total energy is distributed as

$$q\hbar\omega_0 = q_1\hbar\omega_0 + q_2\hbar\omega_0 + q_3\hbar\omega_0$$

or equivalently,

$$q = q_1 + q_2 + q_3, \quad (q_1, q_2, q_3 \geq 0).$$

We can organize the number of possible ways a microstate can be distributed by creating a simple table for each q_i . In general, the **multiplicity** (Ω) of a state with q quanta over N oscillators is given by $(N + q - 1)$ partitions (Fig. 18).

We can visually represent how we organize each partition by considering a number of N bins. For example, suppose we have $q = q_1 + q_2 + q_3 = 6$ quanta with $q_1 = 2$, $q_2 = 1$, and $q_3 = 3$. Then, for $N = 3$ bins, we have 2 quanta in Bin 1, 1 quanta in Bin 2, and 3 quanta in Bin 3.

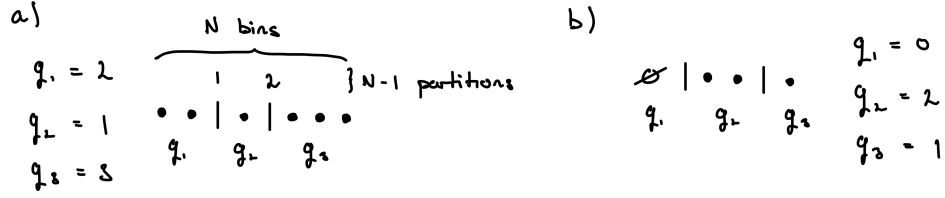


Figure 18: Graphical representation of organizing $N = 3$ bins for $q = 6$. (a) $q_1 = 2$, $q_2 = 1$, and $q_3 = 3$. (b) $q_1 = 0$, $q_2 = 2$, $q_3 = 1$.

The number of possible combinations is such that we organize $q + N - 1$ symbols, of which q symbols are of a given type. Thus, the multiplicity for N bins and q quanta is represented by the formula

$$\Omega(N, q) = \binom{q + N - 1}{q} = \frac{(q + N - 1)!}{q!(N - 1)!}. \quad (49)$$

We call the coefficient $\binom{q + N - 1}{k}$ "n choose k" formula for $n = q + N - 1$ and $k = q$. This counts the different ways we can choose $k = q$ elements of a set of $n = q + N - 1$ elements.

q	q_1	q_2	q_3	Ω
0	0	0	0	1
1	1	0	0	3
	0	1	0	
	0	0	1	
2	2	0	0	6
	1	1	0	
	1	0	1	
	0	2	0	
	0	1	1	
	0	0	2	

Table 4: Possible ways of arranging a microstate for $q = 0, 1, 2$. The multiplicity is given by Ω .

Let us consider a simple example. Suppose we flip a coin n times. We wish to know how many sequences exist where a tail shows up k times. From the binomial expansion, we have that

$$(h + t)^n = \sum_{k=0}^N \binom{n}{k} h^{n-k} t^k.$$

Expanding explicitly, we see that

$$\begin{aligned} (h + t)^1 &= h + t \\ (h + t)^2 &= h^2 + ht + th + t^2 \\ (h + t)^3 &= h^3 + h^2t + t^2h + t^3 \\ (h + t)^4 &= h^4 + \dots \end{aligned}$$

If we look at $(h+t)^2$, we can see that (ht) and (th) are both runs with a single tail. Summing them together, we have $\sum_{k=0}^2 (2,1) = 2$. In fact, we can prove the general formula for (n,k) by the method of mathematical induction. This is a physics course so we will show no proof,³ though the formula is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

for any n, k .

2.1.2 Multiplicity vs. Energy Distribution

Consider a system of two weakly coupled solids. Here, energy is exchanged but the total energy of the system is the sum of the constituent energies. We write the total number of multiplicities for this coupled system by

$$\Omega(N_A, q_A; N_B, q - q_A) = \Omega(N_A, q_A) \Omega(N_B, q - q_A).$$

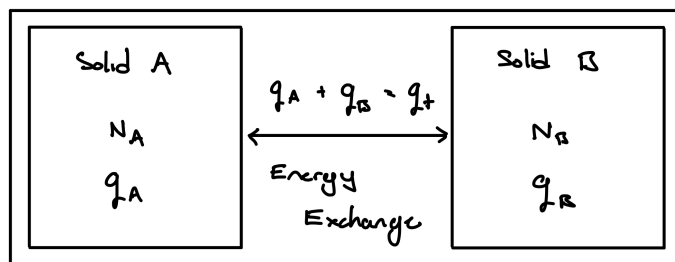


Figure 19: Two Einstein solids that can exchange energy with each other, isolated from the rest of the universe.

It is important we note that in writing this, we make a fundamental assumption about statistical mechanics (and thereby thermodynamics): *all states are equally probable*. Or, in other words, each microstate has an independent chance of occurring.

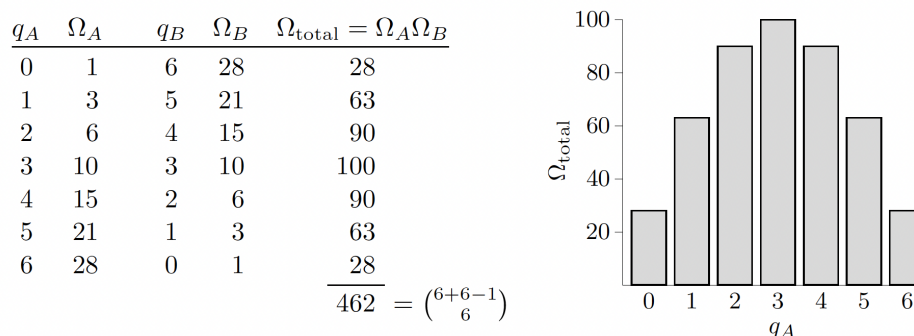


Figure 20: Macrostates and multiplicities of a system of two Einstein solids, each containing three oscillators, sharing a total of six units of energy.

³Maybe I'll do it for practice later when I'm bored.

Sampling over many microstates comes with some larger likelihood of being near the maximum, say $\Omega(q_A)$ if we are interested in the Einstein solid of q_A quanta. So, for a large enough sample size, we may eventually suppose that we satisfy the fundamental assumption that all states are equally probable.

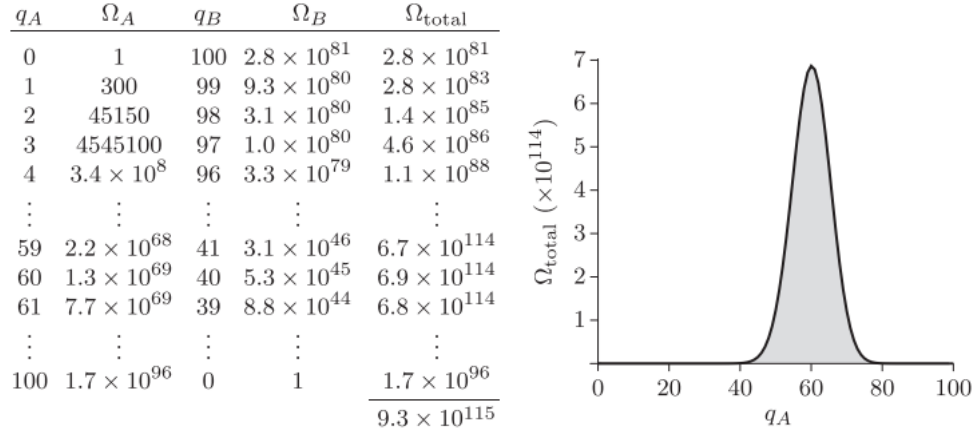


Figure 21: Macrostates and multiplicities of a system of two Einstein solids, with 300 and 200 oscillators respectively, sharing a total of 100 units of energy.

Using the general formula for the multiplicity of a single Einstein solid, we can compute, for example, the total multiplicity of this coupled system. In Fig. 19, we consider the case of $q = 6$ quanta (oscillators), three of which are equally distributed among the solids. By assuming that all states are equally probable, we are able to decompose the total microstates into a product of the microstates of each solid. Indeed, we arrive at $\Omega_{\text{total}} = \Omega_A \Omega_B = 462$.

In Fig. 21 a system of two Einstein solids are graphed with computed multiplicities for the parameters $N_A = 300$, $N_B = 200$, and $q = 100$. Notice that there are 101 possible macrostates with only a few shown in the table. The multiplicities show that even the *least* likely macrostate has a multiplicity of $3 \cdot 10^{81}$, while the *most* likely macrostate ($q_A = 60$) has a multiplicity of $7 \cdot 10^{114}$. We can see that the ratio of the multiplicities suggests that the most likely macrostate is 10^{33} times more probable than the least.

Further, we can see that the total number of microstates for all macrostates is $9 \cdot 10^{115}$, though the probability of finding the system in its most likely macrostate is about 7%. The most likely macrostate occurs at $q_A = 60$, and straying further from this shows that the probability of finding the system in a particular microstate drops off largely.

If we were to, say, begin the system at an initial state with $q_A \ll 60$, we would find that energy has flowed from $B \rightarrow A$ (the system starts with much energy in B). This behaviour is *irreversible*: energy flowed spontaneously from $B \rightarrow A$ but never $A \rightarrow B$, which is actually the physical explanation for *heat*. In essence, heat transfer is *probabilistic*. Indeed, this behaviour is the **second law of thermodynamics**:

The spontaneous flow of energy stops when a system is at, or very near, its most likely macrostate. That is, the macrostate with the greatest multiplicity.

this law is not actually fundamental, but is a very strong statement about the probabilities of a given system.

2.1.3 Stirling's Approximation

Notice, in discussing the Einstein solid, that all multiplicities are generally *very large* numbers. In fact, these multiplicities are to the order of $N!$, with N on the order of Avogadro's number ($N \propto 10^{23}$). We might ask: how large is $N!$ itself? Simply, we know that

$$N! = N(N-1)(N-2) \cdots (2)(1)$$

and so we can apply a rough approximation that

$$N! > \left(\frac{N}{2}\right) \left(\frac{N}{2}\right) \cdots \left(\frac{N}{2}\right) \quad (N/2 \text{ times}).$$

Neglecting the last $N/2$ factors, and taking all $k > N/2$ as $N/2$, we arrive at the approximation

$$N! \approx \left(\frac{N}{2}\right)^{N/2}.$$

A closer approximation we can use is Stirling's approximation, which attempts to approximate $N!$ as

$$N! \approx \left(\frac{N}{e}\right)^N = N^N e^{-N}.$$

However, this approximation is off by a factor of roughly $\sqrt{2\pi N}$. We need only to account for this by multiplying by the lost factor,

$$N! \approx N^N e^{-N} \sqrt{2\pi N}.$$

However, rather than using this direct approximation, we often tend to use logarithms. If we wish to use Stirling's corrected approximation on the log of $N!$, we find the formula

$$\ln(N!) \approx N \ln(N) - N + \frac{1}{2} \ln(2\pi N).$$

Notice that if N is a large number, then $N!$ is very large, and since $\ln(N!) \propto N \ln(N) \gg \ln(2\pi N)/2$, we can generally neglect the last term since it is much smaller than both N and $N \ln(N)$.

Suppose now we wish to determine the multiplicity of a large Einstein solid once more. With the exact formula, we may approximate

$$\Omega(N, q) = \binom{q+N-1}{q} = \frac{(q+N-1)!}{q!(N-1)!} \approx \frac{(q+N)!}{q!N!}$$

which requires only a small correction. If we take the natural log on both sides, we find that by Stirling's approximation,

$$\begin{aligned} \ln(\Omega) &\approx \ln[(q+N)!] - \ln(q!) - \ln(N!) \\ &= (q+N) \ln(q+N) - (q+N) - q \ln(q) - N \ln(N) + (q+N) \\ &= (q+N) \ln(q+N) - q \ln(q) - N \ln(N). \end{aligned}$$

Let us suppose that we are in the limit $q \gg N$ so that there is many more quanta of energy than there are oscillators. Then, our expression of $\ln(q+N)$ turns out to be

$$\ln(q+N) = \ln \left[q \left(1 + \frac{N}{q} \right) \right] = \ln(q) + \ln \left(1 + \frac{N}{q} \right).$$

Making a first order Taylor expansion on $\ln(1 + N/q)$, we have that $\ln(1 + N/q) \approx N/q$ and so for $\ln(\Omega)$ we have

$$\begin{aligned}\ln(\Omega) &\approx (q + N) \left[\ln(q) + \frac{N}{q} \right] - q \ln(q) - N \ln(N) \\ &= q \ln(q) + N \ln(q) + N + \frac{N^2}{q} - q \ln(q) - N \ln(N) \\ &= N \ln\left(\frac{q}{N}\right) + N + \frac{N^2}{q}.\end{aligned}$$

Since q is much larger than N in this case, we may assume $N^2/q \ll N$ to be neglected so that

$$\ln(\Omega) \approx N \ln\left(\frac{q}{N}\right) + N.$$

Taking the exponential on both sides, we recover the original estimation of $\Omega(N, q)$,

$$\Omega(N, q) \approx \left(\frac{qe}{N}\right)^N \quad (q \gg N).$$

2.1.4 Multiplicity of two Coupled Einstein Solids in the High-Energy Limit

Consider now two Einstein solids (A , B) of equal size. Then, $N_A = N_B = N$ and assuming their energies are non-equal, they have quanta q_A and q_B . By the approximation we derived, we find that

$$\Omega(N, q) = \Omega_A \Omega_B \approx \left(\frac{q_A e}{N}\right)^N \left(\frac{q_B e}{N}\right)^N = \left(\frac{e}{N}\right)^{2N} (q_A q_B)^N.$$

The two solids share a total energy $q_A + q_B = q$, and so over time we may assume that there is a energy imbalance

$$q_A = \frac{q}{2} + x, \quad q_B = \frac{q}{2} - x$$

where x is the imbalance parameter. Substituting these assumptions into Ω , we find that

$$\Omega \approx \left(\frac{e}{N}\right)^{2N} \left[\left(\frac{q}{2} - x\right)\left(\frac{q}{2} + x\right)\right]^N = \left(\frac{e}{N}\right)^{2N} \left[\left(\frac{q}{2}\right)^2 - x^2\right]^N.$$

Looking at the complicated term, we see that its natural log is

$$\begin{aligned}\ln \left[\left(\frac{q}{2}\right)^2 - x^2 \right] &= \ln \left[\left(\frac{q}{2}\right)^2 \left(1 - \left(\frac{2x}{q}\right)^2\right) \right] = \ln \left[\left(\frac{q}{2}\right)^2 \right] + \ln \left[1 - \left(\frac{2x}{q}\right)^2 \right] \\ &\approx \ln \left[\left(\frac{q}{2}\right)^2 \right] - \left(\frac{2x}{q}\right)^2.\end{aligned}$$

Thus, we find after exponentiating both sides,

$$\Omega(N, q) = \left(\frac{e}{N}\right)^{2N} \left(\frac{q}{2}\right)^{2N} e^{-N(2x/q)^2} = \Omega_{\max} e^{-x^2/(2\sigma_x^2)}, \quad \Omega_{\max} = \left(\frac{eq}{2N}\right)^{2N}.$$

We have made the substitution that

$$2\sigma_x^2 = \frac{q^2}{4N} \implies \sigma_x = \frac{q}{2\sqrt{2N}}$$

which shows that over a range q , the width of our probability curve is $q/\sqrt{N}$.

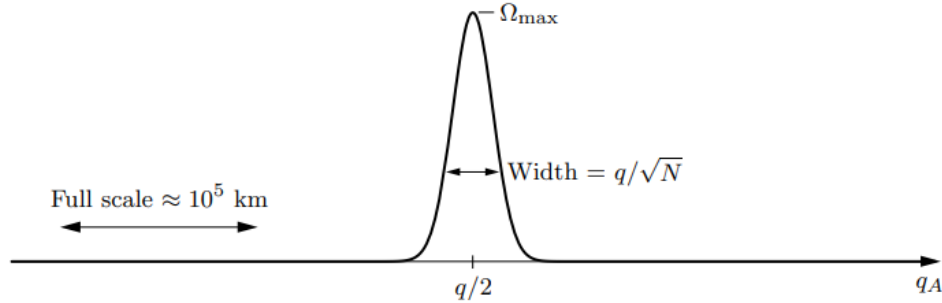


Figure 22: Multiplicity of a system of two large Einstein solids with many quanta per oscillator. We only show a tiny fraction of the full horizontal scale.

In relative terms, this deviation is actually a very small number. Indeed, since $N \propto 10^{23}$, we expect that $1/\sqrt{N} \propto 3 \cdot 10^{-12}$. If we were, funnily enough, to print this graph, multiplying by the width of what the paper size (enough to cover only the curve) would be shows that

$$\frac{10 \text{ cm}}{\sqrt{N}} = 0.3 \text{ pm}$$

which is less than the size of an atom (at about 100 pm). We can see that macrostates are only likely if

$$\frac{q}{2} - \frac{q}{2\sqrt{2N}}(10) < q < q + \frac{q}{2\sqrt{2N}}(1)$$

which is practically only if $q \approx q/2$.

2.1.5 Entropy and Temperature of the Einstein Solid

To summarize our results, we have shown that multiplicity generally has two key properties:

- (1) it is a very large number.
- (2) It increases with both U (energy to distribute) and N (number of constituents for the Einstein solid).

While we have derived these results for the Einstein solid, they hold for any system (with some special exceptions). The multiplicity of two weakly coupled (or weakly interacting) systems A and B , we know distribute multiplicities according to

$$\Omega_{AB}(U_A, U_B) = \Omega_A(U_A)\Omega_B(U_B)$$

where U_A and U_B are the energies held by systems A and B respectively. Correspondingly, we have that the total energy is distributed as

$$U = U_A + U_B$$

which makes use of the specific property of energy additivity (we approximate interaction energies to be negligible). We saw that plotting Ω_{AB} as a function $\Omega_{AB}(U, U_A)$ shows a sharp maximum at U_A (with U fixed). We calculated parameters of Ω_{AB} in energy partitions

$$q_A = \frac{U_A}{\epsilon}, \quad q_B = \frac{U_B}{\epsilon}, \quad q = \frac{U}{\epsilon}$$

where ϵ is the spacing in energy between two levels of a single HO in the Einstein solid. Taking the limit that $U \gg \epsilon$, we can consider U to be a quasi-continuous variable; energy exchanges are relevant at a scale at least

$$\frac{q}{\sqrt{N}}\epsilon \gg \epsilon.$$

Notice that since Ω_{AB} has a maximum at some value, say $\bar{q}_A$, we can equivalently write that

$$\frac{d}{dU_A}\Omega_{AB}(U_A) = 0$$

(we have that U is constant for now and do not specify it any further). It follows that

$$\begin{aligned} \frac{d}{dU_A}\Omega_{AB}(U_A) &= \frac{d}{dU_A}[\Omega_A(U_A)\Omega_B(U - U_A)] \\ &= \Omega_B(U - U_A)\frac{d}{dU_A}\Omega_A(U_A) - \Omega_A(U_A)\frac{d}{dU_B}\Omega_B(U - U_A). \end{aligned}$$

At the maximum point of this curve, we have that

$$\frac{1}{\Omega_A}\frac{d}{dU_A}\Omega_A(U_A) = \frac{1}{\Omega_B}\frac{d}{dU_B}\Omega_B(U_B).$$

Solving this differential equation, we find the solution

$$S = k \ln(\Omega) \tag{50}$$

which we define as the **entropy** of our system.

2.2 The Second Law of Thermodynamics

We discuss in more detail the second law of thermodynamics. If we assume two systems that are free to exchange energy, we can say that they reach a joint macrostate which satisfies the condition

$$\frac{dS}{dU_A} = \frac{dS}{dU_B}.$$

We know from our discussion of thermal equilibrium that temperature is the empirical parameter for two systems. We may propose, in fact, that the relation

$$\frac{1}{T} = \frac{dS}{dU} \tag{51}$$

holds. To see how this is so, we consider the multiplicity of two coupled systems maximized at equilibrium. For any energy U_A , we have that

$$\Omega_{AB}(U_A) \leq \Omega_{AB}(U_{Am})$$

where we let U_{Am} be the most likely macrostate. Correspondingly, we find that the following relation holds.

$$S_{AB}(U_A) < S_{AB}(U_{Am})$$

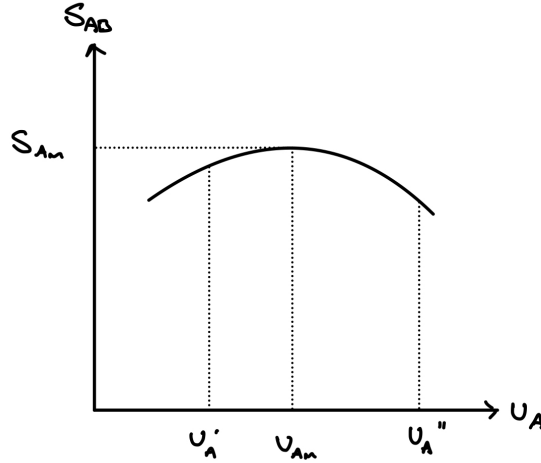


Figure 23: Graphical representation of a potential entropy curve against U_A with values U'_A , U_{Am} , and U''_A .

We know also that if U_A and U_B weakly interact, then we have the relation

$$\ln(\Omega_A \Omega_B) = \ln(\Omega_A) + \ln(\Omega_B).$$

It immediately follows that the equivalent entropy relation holds,

$$S_{AB} = S_A + S_B. \quad (52)$$

Consider now the special case $U_A = U'_A < U_{Am}$. Then, we have that

$$\left. \frac{d}{dU} S_{AB} \right|_{U_A=U'_A} = \left. \frac{dS_A}{dU_A} \right|_{U_A=U'_A} + \left. \frac{dS_B}{dU_A} \right|_{U_A=U'_A} > 0.$$

If we relate, from our previous derivation, the derivative in S_B with respect to U_A by

$$\frac{dS_B}{dU_A} = -\frac{d}{dU_B} S_B(U - U_A),$$

then we find that

$$\left. \frac{dS_A}{dU_A} \right|_{U_A=U'_A} > \left. \frac{d}{dU_B} S_B(U - U_A) \right|_{U_B=U'_B}$$

for $U'_B = U - U'_A$. Indeed, the hypothesis that $1/T = dS/dU$ shows that

$$\frac{1}{T_A} > \frac{1}{T_B} \implies T_A < T_B.$$

If we look at the similarly equivalent case $U_A < U_{Am}$, we can see that A must receive some energy $(U_{Am} - U_A) > 0$, and so B must provide energy for this occur. Empirical observations agree with

our hypothesis for systems that start with $U_A < U_B$ at the initial point.⁴ Thus, we consider the case $U_A = U_A'' > U_{Am}$, so that we find (by similar argument)

$$\frac{d}{dU_A} S_{AB} < 0 \implies T_A > T_B.$$

We have, from empirical observations, that systems not presently in equilibrium evolve from $U_A > U_{Am}$ to $U_A \approx U_{Am}$. That is, system A gives energy away with $T_A > T_B$ as we found earlier. As an important note, we have used (implicitly) that the entropy S_{AB} is a function of U_A , and indeed we assume the maximal value $dS_{AB}/dU_A = 0$. We are interested in whether the temperature is infinite, and we note that $1/T = dS/dU$ assumes $S(U)$ is the entropy of some system of its own energy at equilibrium.

That is, $S_{AB}(U_A)$ is the entropy of the system A - B against the entropy in one of its subsystems. So, we would find that

$$\frac{1}{T} = \frac{dS_{AB}}{dU_A}$$

makes zero sense. The derivative S_{AB} against U_A is **not** a temperature, and we should not expect it to follow that relation.

We will try to make sense of our current discussion graphically. For the Einstein solid, we have that the multiplicity (and hence entropy) increases with energy. In the high energy limit ($q \gg N$), we know that

$$\Omega(N, q) = \left(\frac{eq}{N}\right)^N, \quad q = \frac{U}{\epsilon}$$

holds, where ϵ is some energy partition. We have derived earlier the relation

$$S = k \ln(\Omega) = kN \left[\ln\left(\frac{U}{\epsilon}\right) + 1 - \ln(N) \right]$$

which shows that the energy dependence is logarithmic in high energy. Consider now two systems A and B . After establishing contact, we have by conservation of energy that

$$U = U_A + U_B.$$

Consider the curve U_A first. We can see that the slope of S_A is such that

$$\frac{dS_A}{dU_A} \neq -\frac{dS_B}{dU_A} = \frac{dS_B}{dU_B}.$$

Indeed, we also find that

$$\frac{1}{T_A} \neq \frac{1}{T_B}.$$

Fig. 24 shows that the entropy is a **concave** function of the free parameter U_A . This is, in fact, a general behaviour of entropy for a system where there is freedom to redistribute extensive quantities. This holds for not only energy, but also volume, number of particles, and similar parameters. Naturally, we may ask if there is a maximum that always exists. Indeed, we require that the slope satisfies

$$\frac{1}{T} = \frac{\partial S}{\partial U}$$

⁴Must be confirmed, but is likely true.

where $T \rightarrow 0$ is larger for a system of any size at $U \rightarrow 0$ than for another system of any size at finite U . This is generally satisfied.

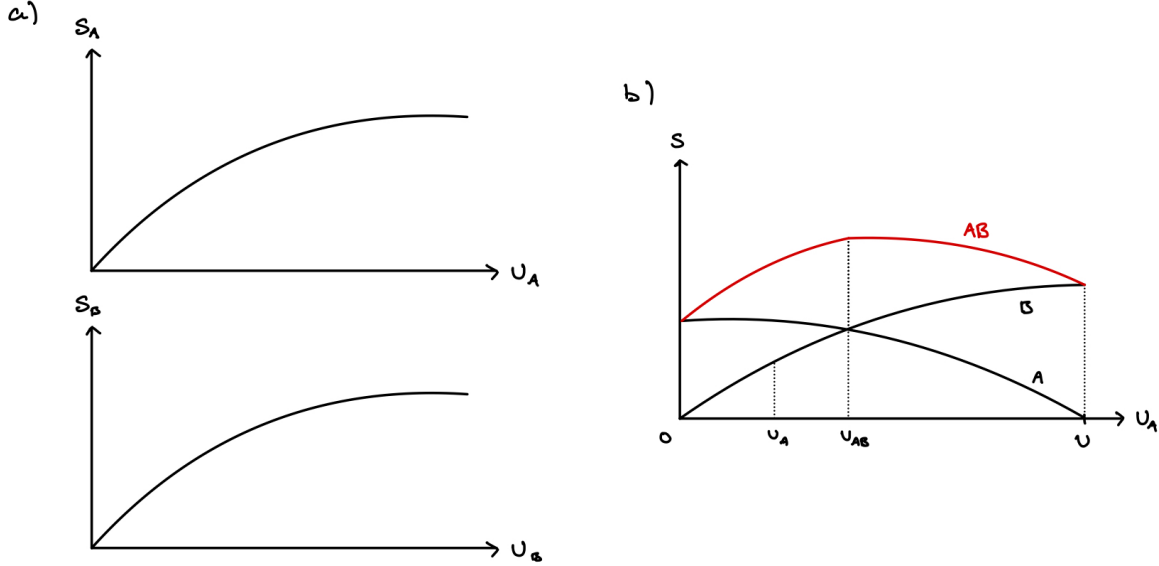


Figure 24: Systems A and B represented graphically. (a) System A and system B plotted against their own energies. (b) System S_{AB} plotted against U_A .

Recalling the entropy relation in the high energy limit, we use the fact that $1/T = \partial S / \partial U$ so that we have

$$\frac{1}{T} = \frac{\partial S}{\partial U} = \frac{kN/\epsilon}{U/\epsilon} = \frac{kN}{U}.$$

Rearranging, we find that $U = kNT$, and so we may define the heat capacity to be

$$C = \frac{\partial U}{\partial T} = kN$$

which is in agreement with the Dulong-Petit law.

2.3 The Ideal Gas Revisited

We will now discuss the connection between microstates and macrostates for an ideal gas. Firstly, we should note how the ideal gas is different from the Einstein solid.

Volume. For the ideal gas, we consider the volume to be a parameter. The number of microstates depends on both energy and volume, and by introducing volume as a parameter we can establish the foundation to link microscopic aspects of the ideal gas and work.

Number of Particles. The number of particles (atomic or molecular) is another important parameter. Entropy (and multiplicity) depends on the number of particles in the ideal gas, which leads into the chemical potential of the gas, another important quantity in thermal physics.

Microstates. The calculation of the number of microstates brings some interesting complications to our system, which involve continuous variables and the definition of energy in that context.

2.3.1 Multiplicity of the Ideal Gas

Consider a monoatomic gases formed of a single atom. For simplicity, we will take the container holding this "single atom gas" to be a parallelepiped with edges L_x , L_y , and L_z , with total volume given by

$$V = L_x L_y L_z.$$

We can classically represent the position by the vector $\mathbf{r} = (x, y, z)$, and the momentum by the vector $\mathbf{p} = (p_x, p_y, p_z)$.

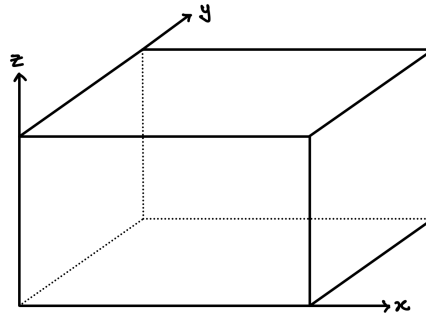


Figure 25: Parallelepiped in Cartesian space. The side lengths (not shown) are L_x , L_y , and L_z respectively.

We know only that the constraints on every position is such that $0 \leq x_i \leq L_i$, where $i = x, y, z$, and that we have the potential energy function

$$U = \frac{|\mathbf{p}|^2}{2m}.$$

However, we still have infinitely many ways we can choose coordinates. Let us consider first one dimension along the interval, say $[0, L_x]$, partitioned into sub-intervals Δx . By Heisenberg's Uncertainty Principle (HUP), it holds that

$$\Delta x \Delta p_x = h.$$

The allowed values of momentum correspond to either two parts on the p_x axis, with values $\pm\sqrt{2mU}$. In general, the energy is not a well-defined quantity, and is only defined within some uncertainty ΔU . So, we have

$$\mathbf{p}(U) = \pm\sqrt{2mU}, \quad \text{range}(\mathbf{p}) = (\mathbf{p} - \Delta\mathbf{p}, \mathbf{p})$$

along with its corresponding range. It follows that the number of allowed states of a particle within the gas is given by

$$\Omega_{1,1} = \left(\frac{L_x}{\Delta x} \right) \left(2 \frac{\Delta \mathbf{p}}{\Delta p_x} \right) = \frac{2L_x}{h} \Delta \mathbf{p}$$

where we define $\Omega_{l,N}$ corresponds to a single state, $l = 1$, and a single dimension, $N = 1$. Consider now an atom in $N = 2$ dimensions.

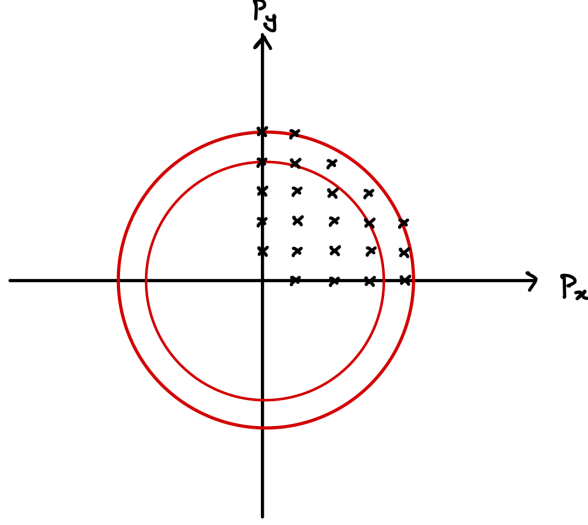


Figure 26: Momentum space representation with axes p_x and p_y . A cross represents a possible microstate (discrete) within constant energy contours (red).

A constant energy contour in Figure 26 is given by

$$p_x^2 + p_y^2 = \sqrt{2mU}$$

and the number of microstates is given by

$$\Omega_{1,2} = \left(\frac{L_x L_y}{\Delta x \Delta y} \right) \left(\frac{2\pi \mathbf{p} \cdot \Delta \mathbf{p}}{\Delta p_x \Delta p_y} \right) = \frac{2\pi \sqrt{2mU} A}{h^2} \Delta \mathbf{p}.$$

Here, we define $A = L_x L_y$ to be the area of the 2-dimensional domain, so that we may condense $\Omega_{2,2}$ into a single form.

Now suppose we have an atom in $N = 3$ dimensions. A single microstate in this case is represented by

$$\Omega_{1,3} = \left(\frac{L_x L_y L_z}{\Delta x \Delta y \Delta z} \right) \left(\frac{\text{spherical shell volume}}{\text{volume of a single element of } \mathbf{r} \text{ space}} \right).$$

A spherical shell volume is given by its 2-dimensional area times its shell thickness, which we may write as

$$\text{spherical shell volume} = \text{2-dimensional area} \times \text{shell thickness} = 4\pi |\mathbf{p}|^2 \Delta \mathbf{p}.$$

A volume in 3-dimensional momentum space, for a single allowed value of momentum, is given by the relation

$$\Delta p_x \Delta p_y \Delta p_z$$

so that we may write

$$\Omega_{1,3} = \frac{(\sqrt{2mU})^2 V}{h^3} \Delta \mathbf{p}.$$

Generalizing to N atoms, we note that there $3N$ spatial coordinates given by x_i , y_i , and z_i respectively, for $i = (1, N)$. This quantization conditions holds for each atom (i.e. x_i can only take values spaced by Δx_i). From HUP, we find that the momentum p_{ix} can only take values separated by

$$\Delta p_{ix} = \frac{h}{\Delta x_i}.$$

This holds for each other momenta in the Cartesian directions. Thus, we fulfill the constraint such that

$$(p_{1x}^2 + p_{1y}^2 + p_{1z}^2) + (p_{2x}^2 + p_{2y}^2 + p_{2z}^2) + \cdots + (p_{Nx}^2 + p_{Ny}^2 + p_{Nz}^2) = 2mU.$$

Notice that this equation is that of a sphere in a $3N$ -dimensional space. The area of such a sphere is given by the generalized formula

$$A_d(r) = \frac{2\pi^{d/2}}{(d/2 - 1)!} r^{d-1}$$

where d corresponds to the dimension of the space. The multiplicity (which is wrong, in fact) follows in the form

$$\Omega_{N,3}^{\text{wrong}} = \left[\frac{L_x L_y L_z}{\Delta x \Delta y \Delta z} \right]^N \left[\frac{A_{3N}(\sqrt{2mU}) \Delta \mathbf{p}}{(\Delta p_x \Delta p_y \Delta p_z)^N} \right] = \frac{V^N 2\pi^{3N/2} (\sqrt{2mU})^{3N-1} \Delta \mathbf{p}}{h^{3N} (3N/2 - 1)!}.$$

We may approximate this result by adjusting the factorial, neglecting $\Delta \mathbf{p}$, etc. That is, we will arrive at the approximation

$$\Omega_{N,3}^{\text{wrong}} \approx \frac{V^N}{h^{3N}} \frac{\pi^{3N/2}}{(3N/2)!} (\sqrt{2mU})^{3N}.$$

Notice that we denoted this equation with "wrong". In fact, during our entire derivation, we have neglected the indistinguishability aspect of the atoms. For example, assigning a position $\mathbf{r}_1$ and $\mathbf{p}_1$ to some atom 1, and a position $\mathbf{r}_2$ and $\mathbf{p}_2$ to some atom 2, has the same result as assigning positions $\mathbf{r}_2$ and $\mathbf{p}_2$ to atom 1 and similar. The problem is that *atoms do not have labels!* Indeed, we have double-counted our configuration. What we have done, essentially, is count each configuration $N!$ times for N atoms.

Our argument would have been correct as long as no two atoms had the same momentum of position. However, this is unlikely, so a small contribution to the sum is for the low density limit. Should we have indistinguishability, we find that our expression is actually of the form

$$\Omega(U, N, V) = \frac{1}{N!} \left(\frac{V}{h^3} \right)^N \frac{(\sqrt{2\pi m U})^{3N}}{(3N/2)!}. \quad (53)$$

In general, the formula for the multiplicity takes on the form

$$\Omega(U, N, V) = f(N) V^N U^{3N/2}$$

where $f(N)$ is some complicated function of N .

2.3.2 Entropy of the Ideal Gas

Notice from our previous result that $\Omega(U, N, V)$ is similar in form to the Einstein solid, where the entropy depends on energy as U^{3N} . Each oscillator in an Einstein solid has two quadratic DoF, plus one quadratic DoF in an ideal gas to make a total of $3N$ DoF. Applying Stirling's approximation on each factorial in our expression for Ω , we find that

$$\Omega(U, N, V) \approx \left(\frac{V}{N}\right)^N \left(\frac{4\pi mU}{3Nh^2}\right)^{3N/2} e^{5N/2}.$$

Thus, we find that the entropy of an ideal gas is given by

$$S = kN \left[\ln \left(\frac{V}{N} \left(\frac{4\pi mU}{3Nh^2} \right)^{3/2} \right) + \frac{5}{2} \right] \quad (54)$$

which is known as the **Sackur-Tetrode Equation**. Suppose we wish to calculate the temperature of the ideal gas. Then, we find from the slope of the entropy, that

$$\frac{1}{T} = \frac{\partial S}{\partial U} = kN \frac{d}{dU} \left[\frac{3}{2} \ln \left(\frac{4\pi mU}{3Nh^2} \right) \right] = \frac{3}{2} kN \left(\frac{3Nh^2}{4\pi mU} \right)^{3/2} \left(\frac{4\pi m}{3Nh^2} \right)^{3/2} U^{1/2} = \frac{3}{2} kNU.$$

Thus,

$$U = \frac{3}{2} kNT. \quad (55)$$

To check if this is in agreement with our equipartition theorem, we need only to recognize that $f = 3$ for three DoF. Notice that in doing so, we assume that the equipartition theorem is valid for an ideal gas. For more complicated additions of DoF, we would have to utilize tools from statistical mechanics, which we are not equipped with yet.

2.3.3 Isoentropic Processes and Free Expansion

Consider the unique case where the entropy is constant. Then, we find that, from the general formula of multiplicity,

$$\frac{S}{k} = k \ln(\Omega) = \ln(f(N)) + N \ln(V) + \frac{3N}{2} \ln(U) \implies \frac{1}{N} \left[\frac{S}{k} - \ln(f(N)) \right] = \ln(VU^{3/2}).$$

Since S is constant (and by extension the entire LHS), taking the exponential shows that

$$VU^{3/2} = \text{constant}.$$

We know that $U \propto T$, so we thus conclude also that

$$VT^{3/2} = \text{constant}.$$

Citing the ideal gas law, we find that

$$V(PV)^{3/2} = P^{3/2}V^{5/2} = \text{constant} \implies (P^{3/2}V^{5/2})^{2/3} = PV^{5/3} = \text{constant}$$

where we have taken the 2/3-rd power on both sides. Notice that this result is consistent with the equation for an adiabatic process, where we had instead

$$PV^\gamma = \text{constant}, \quad \gamma = \frac{f+2}{f} = \frac{5}{3}$$

for a monoatomic gas. We thus conclude that an isentropic process is an adiabatic process, which becomes rather important in the discussion of the thermodynamic identity.

Consider now an ideal gas in a partition within a container. Suppose that we open an orifice within the wall of the partitions so that the gas is free to move into the empty partition. Here, gas atoms pass through and eventually fill the entire container uniformly. The energy of the gas remains constant since the kinetic energy of atoms does not change as they pass through the gap, and so we conclude that there is no work being done. Indeed, the walls are not "displaced", and so the change in volume alone is not equivalent to the work being done. How, then, is work being done?

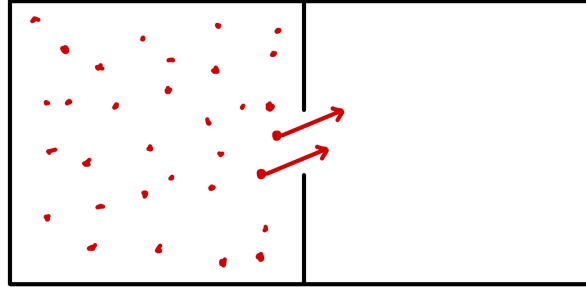


Figure 27: Gas atoms free to pass through an orifice within a partitioned container.

From the first law of thermodynamics, we find that

$$\Delta U = Q + W = 0 \implies Q = 0 \quad (W = 0).$$

Instead, we may ask whether the entropy changes. We note first that the Sackur-Tetrode equation holds for both the initial state and the final state of our gas. So, from our general formula,⁵ we find that

$$\Delta S = k[\ln(\Omega(U, N, V_f)) - \ln(\Omega(U, N, V_i))] = k \ln \left(\frac{f(N) V_f^N U^{3N/2}}{f(N) V_i^N U^{3N/2}} \right) = kN \ln \left(\frac{V_f}{V_i} \right) > 0.$$

Of course, the entropy change is positive since $V_f > V_i$ (the gas takes up more volume, obviously), which is a result that underlines the irreversibility of this process. Indeed, once the atoms fill the entire volume uniformly, the chance that they end up in the initial state is negligibly small.

A slightly more complicated situation is deriving the entropy of mixing. Suppose now we have two gases, namely A and B , in two halves of the same container, each at volumes V with N particles. With an opened partition, both gases end up occupying the full container uniformly. Since entropy is additive, we have that $S = S_A + S_B$ in the initial state and the final state.

However, we should be careful: initially, we may add the entropies based on the total multiplicity being the product of multiplicities of the gases A and B . In the final state, to find the product of multiplicities we require more careful consideration of the system. Entropy additivity still holds, since in this case the gases do not interact with each other and have relatively low density.

The change in entropy follows simply,

$$\Delta S = \Delta S_A + \Delta S_B$$

⁵you may use the Sackur-Tetrode equation instead, the simplification is just more annoying.

and for each gas, we have that

$$\Delta S_{A,B} = kN \ln\left(\frac{V_f}{V_i}\right) = kN \ln\left(\frac{2V_i}{V_i}\right) = kN \ln(2)$$

assuming that the container is partitioned at the centre. Thus, we have

$$\Delta S = 2kN \ln(2).$$

2.3.4 Gibbs Paradox

Suppose now both gases A and B consist of the same atoms. Then, our previous formula for ΔS is simply such that

$$\Delta S = 0.$$

If the entropy of mixing holds, then we naturally may wonder if the reverse holds. Assuming the same setup as before, we can instead remove the partition at the initial state, then at some later time re-add it. Of course, this would imply that

$$\Delta S = -2kN \ln(2) < 0$$

which violates the second law of thermodynamics. This apparent contradiction is linked to the **Gibbs paradox**. To illustrate this, assume that the indistinguishability of atoms was neglected. Then, we expect a multiplicity larger by approximately

$$N! \approx N^N$$

and thus an entropy larger by

$$S \approx kN \ln(N).$$

If we redo the derivation for the Sackur-Tetrode equation, we find instead that we have the entropy

$$S' = kN \left[\ln \left(V \left(\frac{4\pi m U}{3N h^2} \right)^{3/2} \right) + \frac{3}{2} \right].$$

Considering an ideal gas with $2N$ atoms within a volume of $2V$ and possessing energy $2U$, we find that

$$S' \approx 2kN \ln \left(2V \left(\frac{4\pi m 2U}{6N h^2} \right)^{3/2} \right)$$

neglecting the last factor. However, we can also calculate the entropy by summing for both halves of the gas, so that

$$S'' = 2kN \ln \left(V \left(\frac{4\pi m U}{3N h^2} \right)^{3/2} \right) > S'.$$

We have two results. One of entropy $S'' > S'$, and another of entropy just S' . This is the Gibbs paradox.

2.4 The Thermodynamic Identity

Recall the general form for the multiplicity of an monoatomic ideal gas. If we consider a situated where to gases, A and B , are set to be in contact and allowed to exchange energy within a total fixed volume, we have that

$$V = V_A + V_B, \quad U = U_A + U_B$$

where U and V are fixed. We will let the number of particles N_A and N_B be fixed for each gas. Then, the multiplicity for our coupled system is such that

$$\begin{aligned} \Omega(N_A, N_B, U_A, U_B) &= \Omega(N_A, U_A, V_A) \Omega(N_B, U - U_A, V - V_A) \\ &= f(N_A) f(N_B) V_A^{N_A} (V - V_A)^{N_B} U_A^{N_A} (U - U_A)^{N_B}. \end{aligned}$$

Consider the simplest case where $N_A = N_B \equiv N/2$. The volume dependence of Ω arises through the function

$$f_V(V_A) = V_A^{N/2} (V - V_A)^{N/2}$$

and we shall define

$$V_A = \frac{V}{2} + v$$

so that

$$f_V(v) = \left[\frac{V^2}{4} \left(1 + \frac{2v}{V} \right) \left(1 - \frac{2v}{V} \right) \right]^{N/2}.$$

Taking the logarithm, we find

$$\ln(f_V(v)) = N \ln\left(\frac{V}{2}\right) + \frac{N}{2} \ln\left(1 - \left(\frac{2v}{V}\right)^2\right) \approx N \ln\left(\frac{V}{2}\right) - \frac{N}{2} \frac{4v^2}{V^2}.$$

Notice that the multiplicity of Ω depends on volume by the factor

$$e^{\ln(f_v)} \approx \left(\frac{V}{2}\right)^N \exp\left(-\frac{v^2}{2\sigma_v^2}\right), \quad \sigma_v = \frac{V}{2\sqrt{N}} = \frac{1}{2} \sqrt{\frac{V}{\rho}}.$$

For a gas at fixed density, we have that

$$\frac{\sigma_v}{V/2} = \frac{1}{\sqrt{\rho V}}.$$

Further, for a large system, the range of macrostates with a significant probability is negligible relative to the total volume.

2.4.1 Mechanical Equilibrium

Our previous result suggests that equilibrium in two coupled ideal gases results in a well defined value of the volume. This value of the volume corresponds to maximizing the multiplicity and hence the entropy; when the volume of the two subsystems is allowed to change while the total volume remains constant. Indeed,

$$\frac{\partial S_{AB}}{\partial V_A} = 0, \quad S_{AB} = S_A + S_B.$$

Computing the partial derivative, we find

$$\frac{\partial S_{AB}(V_A)}{\partial V_A} = \frac{\partial S_A(V_A)}{\partial V_A} + \frac{\partial S_B(V - V_A)}{\partial V_A} = \frac{\partial S_A(V_A)}{\partial V_A} - \frac{\partial S_B(V - V_A)}{\partial V_B}.$$

At equilibrium, we find that

$$\frac{\partial S_A(V_A)}{\partial V_A} = \frac{\partial S_B(V - V_A)}{\partial V_B}.$$

Notice that in the previous steps we did not specifically include that entropies have other arguments: the energy and the number of particles. However, we have clarified that the derivatives are partial derivatives. The condition that $\partial S_A/\partial V_A = \partial S_B/\partial V_B$ suggests a mechanical equilibrium condition. Naturally, we may ask if $\partial S/\partial V$ is a pressure term. Explicitly checking the units, we find that

$$\left[\frac{\partial S}{\partial V} \right] = [\text{J}][\text{K}]^{-1}[\text{m}]^{-3}$$

which is not [Pa] or similar! In fact, the units of the pascal are such that $[\text{Pa}] = [\text{J}][\text{m}]^{-3}$. In accordance with the units we find, we may infer that the derivative is instead the ratio P/T , so that

$$\frac{P}{T} = \frac{\partial S}{\partial V}. \quad (56)$$

We can verify this relation for the ideal gas. By the Sackur-Tetrode equation, we compute the partial with respect to volume, so that

$$\frac{\partial S}{\partial V} = kN \frac{d}{dV}[\ln(V)] = \frac{kN}{V} = \frac{P}{T}$$

where we have also substituted the ideal gas law, agreeing with our proposed definition.

2.4.2 The Identity

Let us consider now a physical system with a fixed number of particles, N . The entropy in this case depends only on energy and volume, and so we have that $S = S(U, V)$. Implicitly, when we write this type of expression, the entropy is that of a state at equilibrium. Let us consider now how the entropy changes with energy in volume.

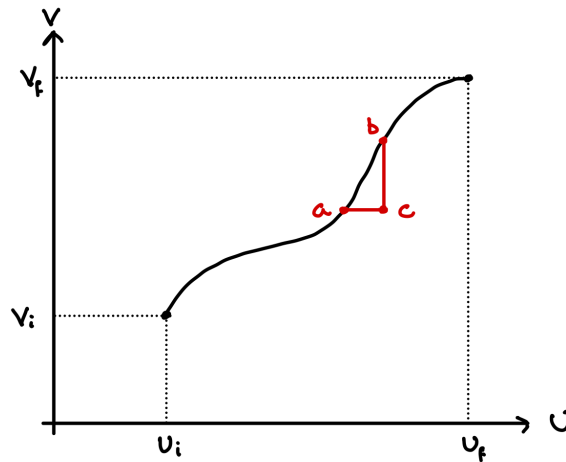


Figure 28: Arbitrary function $V(U)$ across the path $a \rightarrow c \rightarrow b$.

A small change from $a \rightarrow b$ can be written using c as an intermediary, so that

$$\Delta S_{b,a} = \Delta S_{c,a} + \Delta S_{b,c} \approx \frac{\partial S}{\partial U} \Delta U_{c,a} + \frac{\partial S}{\partial V} \Delta V_{b,c} = \frac{\partial S}{\partial U} \Delta U + \frac{\partial S}{\partial V} \Delta V.$$

Note that the entropy is a function of state only, which is what allows us to use the path $a \rightarrow c \rightarrow b$ to calculate the quantities above. Indeed, if we take the limit, we arrive at the differential form of entropy,

$$dS = \frac{\partial S}{\partial U} dU + \frac{\partial S}{\partial V} dV \quad (57)$$

which is the **thermodynamic identity**.

We previously inferred the relation $1/T = \partial S / \partial U$. However, in doing so, we have assumed that $\partial S / \partial U \approx \Delta S / \Delta U$ refers to changes between states that are equilibrium states; specifically, they are states for which the entropy relation $S = k \ln(\Omega)$ holds. When the state of a physical system changes, we require these changes to be slow. In this circumstances, the change in energy is the heat received by the system, so

$$dS = \frac{\partial S}{\partial U} dU = \frac{1}{T} dU = \frac{1}{T} \delta Q \implies \delta Q = T dS.$$

We can rewrite this expression for dS as

$$dU = \left(\frac{\partial S}{\partial U} \right)^{-1} dS - \left(\frac{\partial S}{\partial V} \right) \left(\frac{\partial S}{\partial U} \right)^{-1} dV.$$

The first term is δQ , and so from the first law of thermodynamics, we may conclude that

$$\delta W = - \frac{\partial S}{\partial V} \left(\frac{\partial S}{\partial U} \right)^{-1} dV.$$

We know also that $\delta W = -P dV$, so we might ask if it makes to conclude that

$$P = \frac{\partial S}{\partial V} \left(\frac{\partial S}{\partial U} \right)^{-1}.$$

Indeed, since $\partial S / \partial U = 1/T$ from earlier, we require that

$$\frac{P}{T} = \frac{\partial S}{\partial V}$$

which we have validated earlier for the ideal gas. Here, we have simply arrived at that conclusion from a more general perspective.

2.4.3 State Representations and the Triple Product Rule

We digress temporarily to discuss a potentially confusing issue in thermal physics regarding the changes in state variables. We note that **a state is unique**. However, multiple choices of variables can be used to represent a state. Specifically, in our derivations above we used the energy U and the volume V to specify the state of a system. Instead, we may want to ask if we can use the entropy S and volume V to represent a state. The answer is yes, and doing so results in rather interesting (and useful) properties.

We will discuss these mathematical aspects with a more general notation so that we can focus on the more formal aspects. Consider a function given by $z = z(x, y)$. Here, the variables x and y specify the state, while the value of the function z can be seen as another variable (that depends on x and y) which does not, by itself, specify a state.

Are we able to represent this state in terms of z and y ? If we find an inversion relation between z and x (such that z replaces x), so that we write

$$\tilde{z}_y(x) \equiv z(x, y)$$

which is invertible, then we can replace x by z . However, we require $\tilde{z}_y(x)$ to be a *monotonic* function. It is sufficient that

$$\tilde{z}'_y(x) = \frac{\partial z(x, y)}{\partial x} \neq 0$$

which implies that the derivative, if continuous, is either positive or negative everywhere. Let us write now the new function $x = x(z, y)$ and consider the original function in its differential form,

$$dz = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy.$$

We use a subscript to indicate which other variable is used to condense notation. Suppose z is constant, so that we find

$$\left(\frac{\partial z}{\partial x} \right)_y \frac{dx}{dy} = - \left(\frac{\partial z}{\partial y} \right)_x.$$

Here, the derivative dx/dy has a very specific meaning. Then, taking z as a variable, we may note that x or y are additional variables. Indeed, since we have written that $\tilde{z}_y(x)$ is invertible, we may write

$$\frac{dx}{dy} = \left(\frac{\partial x}{\partial y} \right)_z, \quad \left(\frac{\partial z}{\partial y} \right)_x = \left[\left(\frac{\partial y}{\partial z} \right)_x \right]^{-1}.$$

The second result is based on the notion of inversion which was discussed above. Using either $z(x, y)$ or $y(z, x)$ (such that y replaces z), we may find the **triple product rule**,

$$\left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial z} \right)_x \left(\frac{\partial z}{\partial x} \right)_y = -1. \quad (58)$$

We can gain some intuition on the presence of the negative sign by considering some function of the form

$$z(x, y) = \alpha_x x + \alpha_y y, \quad \alpha_x, \alpha_y > 0.$$

Consider a line of constant z . In the xy -plane, the line has negative slope, while in the other planes the slope is positive. Thus,

$$\left(\frac{\partial x}{\partial y} \right)_z < 0, \quad \left(\frac{\partial z}{\partial x} \right)_y = \alpha_x > 0, \quad \left(\frac{\partial z}{\partial y} \right)_x = \alpha_y > 0.$$

We wish to calculate the quantity $(\partial S / \partial V)_U$ and to do so we make use of the triple product rule. In doing so, we find

$$\left(\frac{\partial S}{\partial V} \right)_U \left(\frac{\partial V}{\partial U} \right)_S \left(\frac{\partial U}{\partial S} \right)_V = -1 \implies \left(\frac{\partial S}{\partial V} \right)_U = - \left(\frac{\partial U}{\partial V} \right)_S \left(\frac{\partial S}{\partial U} \right)_V.$$

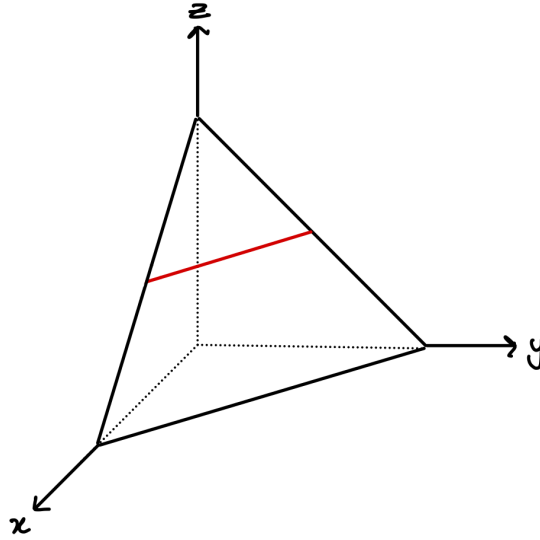


Figure 29: Arbitrary function $z(x, y)$ with a specific line at constant z (in red).

Indeed, we find that

$$\left(\frac{\partial S}{\partial U}\right)_V = \frac{1}{T}.$$

We can associate the quantity $-(\partial U/\partial V)_S$ with P . Indeed, for an adiabatic process, we have the special circumstance where $\delta Q = 0$ and $S = \text{constant}$. Thus, the first law of thermodynamics shows that

$$dU = \left(\frac{\partial U}{\partial V}\right) dV = -P dV.$$

Using this, we arrive at the completed form of the thermodynamic identity, such that

$$dU = T dS - P dV. \quad (59)$$

This identity is effectively a re-statement of the first law and *holds for quasistatic processes*. Quasistatic processes are irreversible, which is implicit in the way we write the thermodynamic identity. Consider, specifically, a small change in the state of the system from $A \rightarrow B$. We assume that such a state can be carried out in the reverse, and so we have

$$\Delta S_{A \rightarrow B} = -\Delta S_{B \rightarrow A} = S_B - S_A.$$

In particular, this process requires that

$$Q_{A \rightarrow B} = -Q_{B \rightarrow A}$$

which does *not* hold for non-quasistatic processes.

2.4.4 Non-Quasistatic Processes

In a non-quasistatic or irreversible processes, we have the condition that

$$dS > \frac{dQ}{T} \implies dU = \delta Q + \delta W < T dS + \delta W.$$

Recall that we may write $\delta W = -P dV$, which we must treat now with some caution. In particular, in a non-equilibrium setting, the pressure of the system may not be well-defined. It is more appropriate to consider the pressure, P , to be that of an external agent performing the work. Considering this, we instead must write

$$dU \leq T dS - P dV$$

which is known as the **thermodynamic inequality**. The special case of equality holds for quasi-static or irreversible processes. In further discussions we will help shade some light on the meaning of the inequality

$$dS \geq \frac{\delta Q}{T}.$$

Free Expansion. The free expansion of an ideal gas was discussed in a previous section. Here, we have $\Delta U = W = 0$, implying that we have also $Q = 0$. The entropy of the gas increases substantially, and so we have $\Delta S > 0$, an irreversible process.

Gas Mixing. Related to free expansion is also the mixing of different gases. Here, the heat involved in the process is $Q = 0$ while $\Delta S > 0$. We may not have $\Delta U = W = 0$ in this case.

2.4.5 Heat Transfer of Two Thermally Coupled Systems

Consider two systems, A and B , coupled by a weak thermal link. The link allows heat to flow, but only very slowly, so that at each moment in time systems A and B are close to equilibrium.

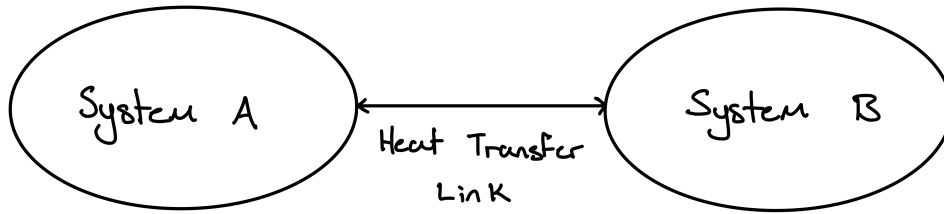


Figure 30: Systems A and B linked thermally.

The two systems are initially at temperatures T_{A_i} and T_{B_i} until they reach equilibrium. At this point, each system is nearly in equilibrium, and so $dS = \delta Q/T$ applies. Taking the integral, we have

$$\Delta S_A = - \int \frac{dQ}{T_A}, \quad \Delta S_B = \int \frac{dQ}{T_B}.$$

Here, dQ is the heat flowing from $A \rightarrow B$. To get the change in entropy of the whole system, we need only to utilize the additivity of entropy,

$$\Delta S_{AB} = \Delta S_A + \Delta S_B = \int \left(\frac{1}{T_B} - \frac{1}{T_A} \right) dQ > 0$$

which is positive at all times, imposing that $T_B < T_A$ at all times. As a whole, the combined system does not receive any heat overall, and so we have $Q_{AB} = 0$, though the entropy increases.

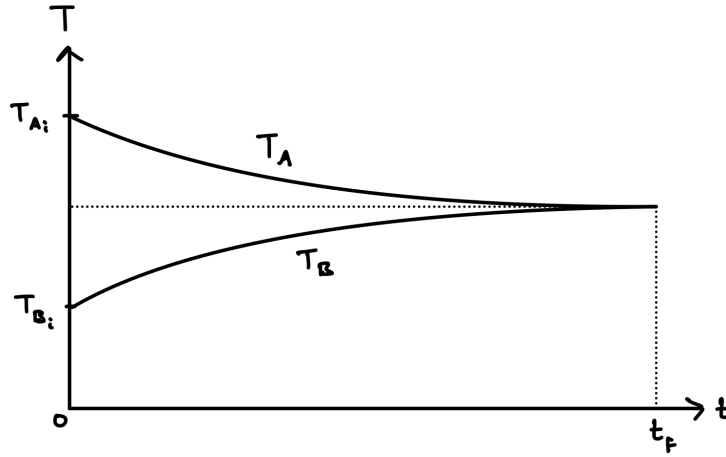


Figure 31: Systems A and B at some initial temperatures T_{A_i} and T_{B_i} reaching equilibrium at a later time t_f .

Slow Thermalization. Consider now the case where systems A and B are now subsystems and weakly connected. The weakly connected subsystem is strongly linked to the environment.

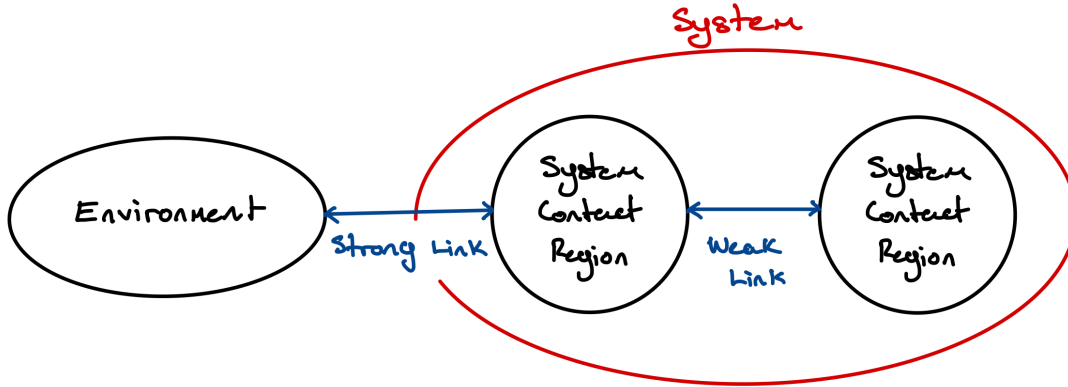


Figure 32: Subsystems A and B contained entirely within their own system connected by a weak link. The system is strongly connected to the environment.

Heat transfer takes place effectively to a contact region. When the environment temperature, say T_e , and contact region, T_{sc} , are nearly equal, we may write the approximation

$$dQ \approx T_e dS \approx T_{sc} dS.$$

However, the interior regions are lagging and internal distribution of heat leads to an additional generalization of entropy (which is discussed in the start of this subsection). We illustrate, with this example, how the thermalization of a system can generate entropy in addition to the minimum entropy determined by $dS_{\min} = dQ/T$, where dQ is the infinitesimal heat transferred from an external system.

Non-quasistatic Work. In comparison of an ideal gas, it is unavoidable that the externally applied pressure exceeds the pressure that corresponds to the ideal gas equilibrium pressure for the volume

$$P_e > P(V).$$

Given that we have the work is equal to the pressure, we may write

$$\delta W = P_e(-dV) > P(-dV),$$

which shows that an overall change between two states has more of its own share of work,

$$W > W_q.$$

Indeed, W_q is the work done in quasistatic conditions. From the first law, we can write the change in quasistatic energy (and additionally the normal change in energy) such that

$$\Delta U = W_q + Q_q, \quad \Delta U = W + Q$$

where Q_q is the quasistatic heat. Thus, we may conclude that

$$W > W_q \implies Q < Q_q = T\Delta S \implies \Delta S > \frac{T}{Q}.$$

2.4.6 The Chemical Potential

Consider now two gases that occupy volumes V_A and V_B , but are free to exchange particles and energy.

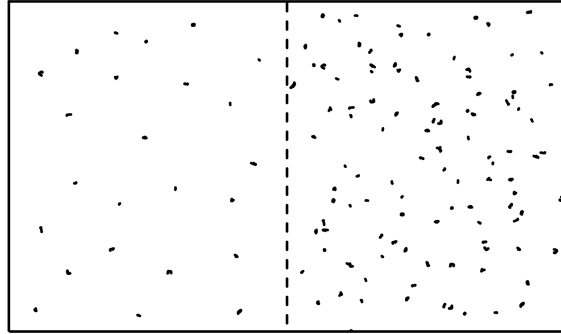


Figure 33: Two gases in a volume V_A and V_B , free to exchange particles by passing through the dashed line.

The total entropy, we may write, is

$$S_{AB}(\{V\}, \{N\}, \{U\}) = S_A(V_A, N_A, U_A) + S_B(V_B, N - N_A, U - U_A)$$

where we write $\{V\}$, $\{N\}$, and $\{U\}$ to be the state variables $\{V\} = \{V_A, V_B\}$, etc. At equilibrium, the entropy reaches a maximum, and so using similar arguments to analyze thermal and mechanical equilibrium, the following conditions hold

$$\begin{aligned} \frac{\partial S_A}{\partial U_A}(V_A, N_A, U_A) &= \frac{\partial S_B}{\partial U_B}(V_B, N - N_A, U - U_A) \\ \frac{\partial S_A}{\partial U_A}(V_A, N_A, U_A) &= \frac{\partial S_B}{\partial U_B}(V_B, N - N_A, U - U_A). \end{aligned}$$

The first condition is that of thermal equilibrium and corresponds to equality of temperatures. The second condition introduces another quantity, μ , with the condition

$$\frac{\mu}{T} = -\frac{\partial S}{\partial N} \quad (60)$$

for diffusive equilibrium. Indeed, at equilibrium we may have $T_A = T_B$ and $\mu_A = \mu_B$, and so we can write the number of particles to be a variable. In this way, we arrive at a more general thermodynamic identity, in the form

$$dS = \left(\frac{\partial S}{\partial U}\right)_{N,V} dU + \left(\frac{\partial S}{\partial V}\right)_{N,U} dV + \left(\frac{\partial S}{\partial N}\right)_{U,V} dN = \frac{1}{T} dU + \frac{P}{T} dV - \frac{\mu}{T} dN.$$

Multiplying by T , we thus find

$$dU = T dS - P dV + \mu dN, \quad \mu = \left(\frac{\partial U}{\partial N}\right)_{S,V} \quad (61)$$

which is the **thermodynamic identity for the chemical potential**. The diffusive equilibrium condition examined earlier specifically led to

$$\mu = T \left(-\frac{\partial S}{\partial N}\right)_{U,V} = -\left(\frac{\partial S}{\partial U}\right)_{V,N}^{-1} \left(\frac{\partial S}{\partial N}\right)_{U,V}$$

which is consistent with the triple product rule. Here, state variables U , S , and N are involved, and U is kept constant. For fixed S and V , we have the special case

$$dU = \mu dN \quad (S, V = \text{const.})$$

which suggests that the chemical potential is a positive quantity, as increasing the number of units increases the energy. This, however, is **not true**. Keeping the entropy constant, an increase in N must be accompanied by a decrease in energy.

Chemical Potential for an Ideal Gas. For a monoatomic gas, the Sackur-Tetrode equation shows that we can write μ as

$$\begin{aligned} \mu &= -T \frac{\partial S}{\partial N} = -kT \left\{ \ln \left(V \left(\frac{4\pi m U}{3h^2} \right)^{3/2} \right) - \ln(N^{5/2}) + \frac{5}{2} - \frac{5}{2} \right\} \\ &= -kT \ln \left[\frac{V}{N} \left(\frac{4\pi m U}{3Nh^2} \right)^{3/2} \right] \\ &= -kT \ln \left[\frac{V}{N} \left(\frac{2\pi m kT}{h^2} \right)^{3/2} \right]. \end{aligned}$$

where $U = 3kNT/2$. Note that an increase in density leads to an increase in chemical potential, which agrees with the expectation of homogeneity.

Multiple Gases/Liquids. In the special case where we have a system with a mixture of different chemical species, diffusive equilibrium conditions apply to each species. So, for any species denoted by α , the chemical potential is given by

$$\mu_\alpha = -T \left(\frac{\partial S}{\partial N_\alpha} \right)_{U, V, N_i \neq \alpha}. \quad (62)$$

Here, the generalized thermodynamic identity becomes

$$dU = T dS - P dV + \sum_{\alpha=1}^k \mu_\alpha dN_\alpha. \quad (63)$$

In chemistry, chemical potentials are defined relative to the number of moles, so we instead write

$$\mu' = -T \left(\frac{\partial S}{\partial n} \right).$$

2.5 The Paramagnet

Currently we have used two important model systems: the Einstein solid and the ideal gas, which we have used to develop the concept of entropy and validate the expressions of temperature and pressure, relevant to empirical observations (i.e., the Dulong-Petit law and the ideal gas law).

We will now discuss the paramagnet. This system, while less complex than our other model systems (since it has no volume and a simple structure), displays properties with respect to how entropy changes with energy that is rather unintuitive. Indeed, the paramagnet is best explained with quantum natures with a bound energy spectrum.

Classically, we can model the paramagnet with magnetic dipoles, with a magnetic moment μ in an externally applied field, $\mathbf{B}$. The energy of the dipole is given by

$$E = -\mu \cdot \mathbf{B} \quad (64)$$

which is minimum when μ is aligned with $\mathbf{B}$. In the classical picture, a torque acts on the dipole given by

$$\tau = \mu \times \mathbf{B}$$

which tends to align the dipole with the magnetic field.

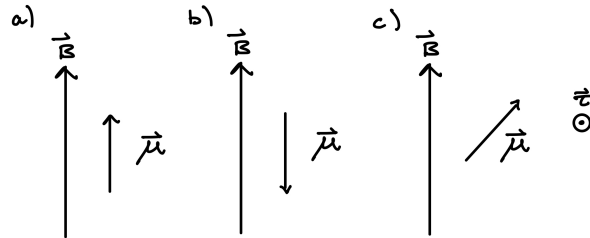


Figure 34: Possible configuration of the classical magnetic dipole. (a) The lowest energy is when $\mathbf{B}$ and μ are aligned. (b) The highest energy is when $\mathbf{B}$ and μ are completely opposite. (c) The case where a torque τ is applied to the dipole system.

However, we will work with a quantum mechanical model of the simplest dipole, the spin 1/2 system. This system has only two allowed values of angular momentum, given by $\hbar/2$ and $-\hbar/2$. The corresponding values of the magnetic moment are denoted by $+\mu$ and $-\mu$, where a single dipole has energy levels $E = \mu B$ or $E = -\mu B$, for the "down" and "up" orientation correspondingly. We should be careful to note that the "down" configuration corresponds to a higher energy state, and vice versa.

Taking N spins, so that

$$N = N_{\uparrow} + N_{\downarrow}$$

we have the total energy to be

$$U = \mu B(N_{\downarrow}) - \mu B(N_{\uparrow}) = \mu B(N_{\downarrow} - N_{\uparrow})$$

where $N_{\downarrow}(N_{\uparrow})$ is the number of down ($U_{\downarrow}$) spins. Neglecting the interaction between spins, we write more explicitly by the relation for N spins,

$$U = \mu B(N - 2N_{\uparrow})$$



Figure 35: The two possible configurations of the spin 1/2 dipole system.

We thus find that the total magnetization is given by

$$M = \mu N_{\uparrow} - \mu N_{\downarrow} = \mu(N_{\uparrow} - N_{\downarrow}) = \mu \left(-\frac{U}{\mu B} \right) = -\frac{U}{B}.$$

A given macrostate of the system is described by U . We should also note that specifying the energy is equivalent to specifying the magnetization, M , or $N_{\uparrow}$. Using this, the multiplicity is given by the n choose k formula,

$$\Omega(N_{\uparrow}) = \binom{N}{N_{\uparrow}} = \frac{N!}{N_{\uparrow}!N_{\downarrow}!}$$

which is the relation we have derived previously. Indeed, we are looking at the number of ways to have $N_{\uparrow}$ spins point up out of a total of N spins. The multiplicity and entropy against the energy have a non-monotonic behaviour, which is in stark contrast to Einstein solids or the ideal gas.

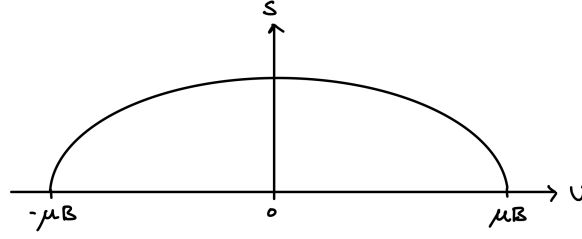


Figure 36: Entropy against the energy with the two possible energy configurations plotted on the energy axis.

Notice, for $-\mu B \leq U < 0$, we have that $\partial S/\partial U > 0$, suggesting that the system increases its entropy by absorbing energy. This is the *expected* behaviour of our system. In the more unintuitive case, we have for $0 < U \leq \mu B$, that $\partial S/\partial U < 0$, suggesting that adding more energy to the system leads to a decrease in energy.

To understand how the system dynamics work, we will consider the thermal dynamics of a paramagnet, labelled with the index p . Putting this paramagnet in contact with an additional arbitrary system, labelled with the index x , we may write that the total energy and entropy are of the form

$$\begin{aligned} U &= U_p + U_x \\ S &= S_p + S_x. \end{aligned}$$

The peculiar nature of the paramagnet does not change the properties of additivity for energy and entropy, so long as the p - x coupling is weak. The combined system tends to maximize its entropy, and so we can calculate the change in entropy as

$$\Delta S = \frac{\partial S_p}{\partial U_p}(U_p)\Delta U_p - \frac{\partial S_x}{\partial U_x}(U - U_p)\Delta U_p = \left[\frac{\partial S_p}{\partial U_p}(U_p) - \frac{\partial S_x}{\partial U_x}(U - U_p) \right] \Delta U_p.$$

We will consider the case where $U_p < 0$. Then, we find that

$$\frac{1}{T_p} \equiv \frac{\partial S_p}{\partial U_p} > 0$$

and so the temperature has the typical positive value of the form

$$\Delta S = \left(\frac{1}{T_p} - \frac{1}{T_x} \right) \Delta U_p > 0.$$

If, for example, we have that $T_p > T_x$, we will find that $\Delta U_p < 0$, and so the paramagnetic gives energy to the x system, which is colder. Similarly, if $T_p < T_x$, the transfer takes place in the opposite direction, so that $\Delta U_p > 0$.

However, while U spans the interval $(-\mu BN, 0)$, the paramagnetic temperature spans the range $(0, \infty)$. A paramagnetic can thermalize with any physical system that has a positive temperature, so long as the additional physical system can give away or take in energy up to μBN .

Consider now the case where the initial paramagnetic energy is $U_p > 0$. Then, if the x -system is normal ($T_x > 0$), the only possibility for the system to evolve is when $\Delta U_p < 0$, since

$$\frac{1}{T_p} \equiv \frac{\partial S_p}{\partial U_p} < 0 \implies \frac{1}{T_x} > 0.$$

Thus, the paramagnetic temperature over the full range has (qualitatively) the a form similar to the $-1/x$ dependence.

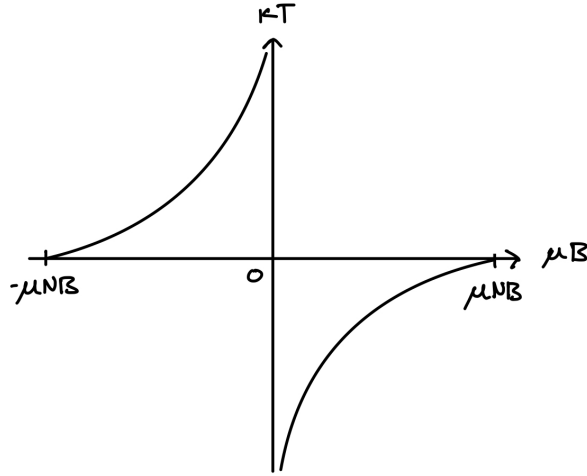


Figure 37: Temperature as a function of energy over the range $(-\mu BN, \mu BN)$ for the μB (energy) axis, and $(-\infty, \infty)$ for the temperature axis.

Of course, we should be careful to note that this dependence is *ideal*. In reality, the temperature will not reach infinity or negative infinity, and is at most limited by $-\mu BN$ or μBN .

MAYBE ADD THE VALUE OF THE FINITE TEMPERATURE AT μBN

As U_p decreases, the temperature crosses from negative to positive, and eventually the paramagnet thermalizes when $T_p = T_x$. These considerations show that establishing thermal contact between a paramagnet and a normal system cannot possibly lead to a paramagnetic increasing its energy, U , to positive values, where more than half the dipoles are in the high energy downs state.

We might ask if one can achieve a state of negative temperature. We can do this by allowing spins to thermalize at a magnetic field that is large enough, followed by a sudden reversal of the magnetic field. Experimentally, this was achieve by Parcell & Pound in 1991, using a set of nuclear spins. In this system, spins interact more strongly with each other than with the solid lattice, leading to equilibrium of the spin system. Eventually, these spins give energy away and end up at a positive temperature $T_p > 0$ and an energy $U_p < 0$.

2.5.1 Analytical Results

By Stirling's approximation, we may write the entropy as

$$\frac{S}{K} \approx N \ln(N) - N_{\uparrow} \ln(N_{\uparrow}) - (N - N_{\uparrow}) \ln(N - N_{\uparrow}).$$

Then, the inverse temperature (by the chain rule), is

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N,B} = \frac{\partial S}{\partial N_{\uparrow}} \frac{\partial N_{\uparrow}}{\partial U}.$$

We know the energy is of the form

$$U = \mu B(N - 2N_{\uparrow}) \implies \frac{\partial N_{\uparrow}}{\partial U} = -\frac{1}{2\mu B}$$

and so taking the derivative of entropy with respect to the number of up states,

$$\frac{\partial(S/K)}{\partial N_{\uparrow}} = -N_{\uparrow} \frac{1}{N_{\uparrow}} - \ln(N_{\uparrow}) + (N - N_{\uparrow}) \frac{1}{(N - N_{\uparrow})} + \ln(N - N_{\uparrow}) = \ln\left(\frac{N_{\downarrow}}{N_{\uparrow}}\right)$$

Indeed, we thus have that

$$N_{\uparrow} = \frac{1}{2} \left(N - \frac{U}{\mu B} \right), \quad N_{\downarrow} = \frac{1}{2} \left(N + \frac{U}{\mu B} \right)$$

so that

$$\frac{1}{T} = \frac{k}{2\mu B} \ln\left(\frac{N - U/\mu B}{N + U/\mu B}\right).$$

Solving this equation for $U(T)$, we find

$$\frac{N - U/\mu B}{N + U/\mu B} = \exp\left(\frac{2\mu B}{kT}\right) \implies U(T) = \mu B N \frac{1 - \exp(2\mu B/kT)}{1 + \exp(2\mu B/kT)} = -\mu B N \tanh\left(\frac{\mu B}{kT}\right). \quad (65)$$

Thus, the magnetization is

$$M = -\frac{U}{B} = \mu N \tanh\left(\frac{\mu B}{kT}\right). \quad (66)$$

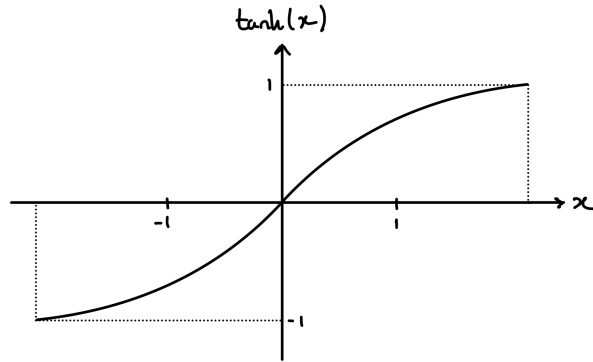


Figure 38: Graph of $\tanh(x)$. Here, $x = \mu B/kT$.

Notice that the magnetization, M , approaches $\mu B N$ when $\mu B/kT \gg 1$. The specific heat at constant B , we find is

$$C_B = \left(\frac{\partial U}{\partial T} \right)_{N,B} = (\mu N B) \frac{\mu B}{kT^2} \frac{1}{\cosh^2(\mu B/kT)} = kN \frac{(\mu B/kT)^2}{\cosh^2(\mu B/kT)}. \quad (67)$$

For electron spins the magnetic moment is the Bohr magneton,

$$\mu_B = eh/4\pi m_e \approx 9.7 \cdot 10^{-24} \text{ J/T} \approx 5.78 \cdot 10^{-5} \text{ eV/T}.$$

For concreteness, we will suppose that $B = 1$ T to represent a strong laboratory field, so that $\mu B \approx 5.810^{-5}$ eV. Thus, at room temperature, we have $kT = 1/40$ eV $^{-1}$. However, in general $\mu B/kT \ll 1$ unless we work at very low temperatures in the Kelvin range. In this range, we have

$$M = \frac{\mu^2 NB}{kT}$$

which is **Curie's law**.

3 Engines and Refrigerators

3.1 Heat Engines

A **heat engine** is a device which can convert heat into useful mechanical work and plays a very important role in thermal physics. Typically a heat engine (and a refrigerator, which we will see later) constitute one of the most important applications in thermal physics. One recognizable type of heat engine is the *steam turbine*, which produces electricity in most of today's power plants. In the same vein, the internal combustion engine used in automobiles can also be describe by a heat engine model.

A heat engine interacts with a hot and cold reservoir to produce mechanical work as shown in Figure 39.

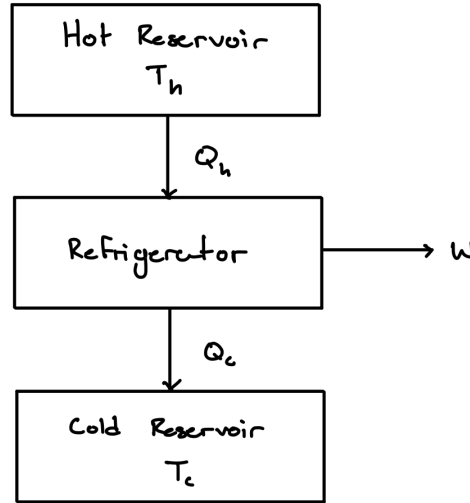


Figure 39: Energy flow diagram for a heat engine.

In the discussion of heat engines, we are interested in the **cyclic engine**, where the state of the system changes periodically while keeping access to heat reservoirs and work sources. Generally, any practical system must run in some form of cycle. From the first law of thermodynamics, we may write that

$$Q_h = Q_c + W$$

where Q_h is the heat flowing from a hot reservoir to the system, Q_c is the heat flowing from the system to the cold reservoir, and W is the work done by the system on some external system. We should note that we work with strictly positive quantities. That is,

$$Q_h, Q_c, W > 0, \quad \forall t \in \mathbb{R}$$

supposing each quantity may be time dependent. The non-time dependent case is trivial. Further, by the second law of thermodynamics, we may also write

$$\Delta S_h + \Delta S_c + \Delta S_{\text{eng}} \geq 0.$$

In particular, we have that $\Delta S_h = -Q_h/T_h$ and $\Delta S_c = Q_c/T_c$. Assuming that the reservoirs are large enough so that a transfer of heat can be considered quasistatic and the temperatures remain constant, we have the special case

$$\Delta S_{\text{eng.}} = 0.$$

In particular, we can assume this is the case since the system's initial and final states are the same over an entire cycle. Thus, we may write

$$-\frac{Q_h}{T_h} + \frac{Q_c}{T_c} \geq 0.$$

The **efficiency** of an engine is written as

$$e = \frac{\text{benefit}}{\text{cost}} = \frac{W}{Q_h}. \quad (68)$$

In the case of our simple heat engine, we have

$$e = \frac{Q_h - Q_c}{Q_h} = 1 - \frac{Q_c}{Q_h} \leq 1 - \frac{T_c}{T_h} = e_{\text{max}}$$

which is less than the **maximum efficiency** by our inequality relations. Notice that the maximum efficiency improves as T_c/T_h decreases.

A thermal engine may be below the maximum value due to two main types of inefficiencies.

- (1) If friction is involved, heat is generated. This extra heat becomes dumped into the cold reservoir, and as a result, reduces the produced work.
- (2) Transfer of heat from the hot reservoir at the temperature T_h and the engine at a temperature $T'_h \lesssim T_h$ leads to an additional production of entropy, written as $\Delta S'$.

Analyzing the second case, we need only to modify our statement on the second law of thermodynamics,

$$-\frac{Q_h}{T_h} + \frac{Q_c}{T_c} \geq \Delta S' \implies -\frac{Q_c}{Q_h} \leq -\left(\frac{T_c}{T_h} + S_{\text{th}} \frac{T_c}{Q_h}\right).$$

An ideal engine requires either the case $T_c \rightarrow 0$ or $T_h \rightarrow \infty$, which are impossible in practice. Practical heat engines must use ambient temperature and cold reservoirs, and a hot reservoir whose temperature is limited only to hundreds of Kelvin.

3.1.1 The Carnot Cycle

The Carnot cycle is an idealized cycle. In the Carnot cycle, heat is absorbed infinitesimally slowly from a hot reservoir at a temperature T_h , while the system is nearly at the same temperature to avoid heat production. Similarly, heat is yielded to the cold reservoir with the system only slightly hotter than T_c .

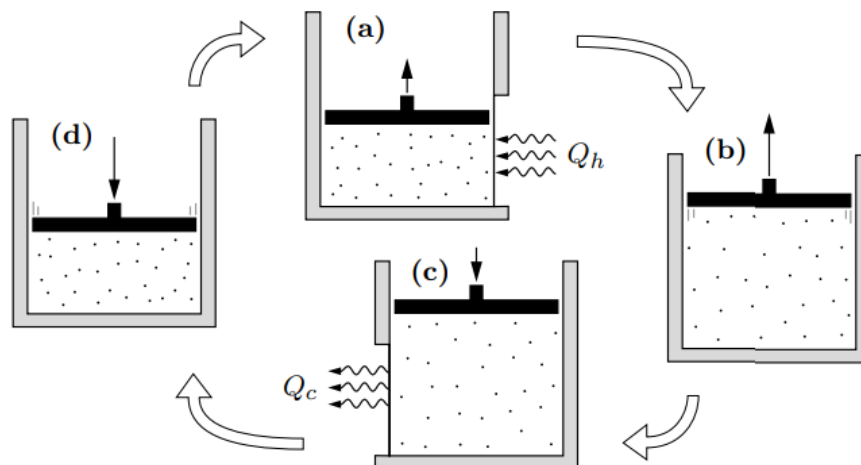


Figure 40: Four steps of a Carnot cycle: (a) isothermal expansion at T_h while absorbing heat; (b) adiabatic expansion to T_c ; (c) isothermal compression at T_c while expelling heat; and (d) adiabatic compression back to T_h . The system must be put into thermal contact with the hot reservoir during step (a) and with the cold reservoir during step (c).

An engine based on any working substance that follows the Carnot cycle reaches the ultimate efficiency

$$e_{\max} = 1 - \frac{T_c}{T_h}.$$

This result can be explicitly confirmed when considering an ideal gas since the heat and work can be calculated explicitly.

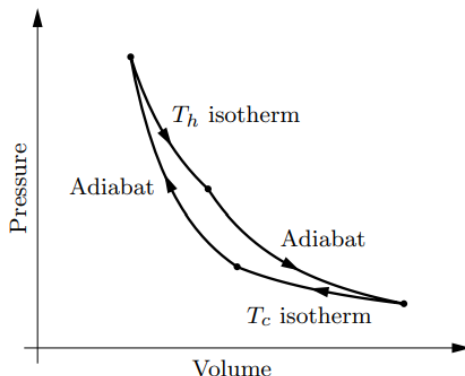


Figure 41: PV diagram for an ideal monoatomic gas undergoing a Carnot cycle.

3.1.2 The Otto Engine

The Otto engine is the engine used in car gasoline engines. It consists of five parts: (1) gas formed of air and gasoline is prepared; (1-2) compression of the gas; (2-3) ignition of the gas to produce heat by chemical processes; (3-4) expansion of the gas which is approximated as adiabatic; (4-1) expelling of the gas mixture to be replaced by a fresh mix, approximated as a constant volume process.

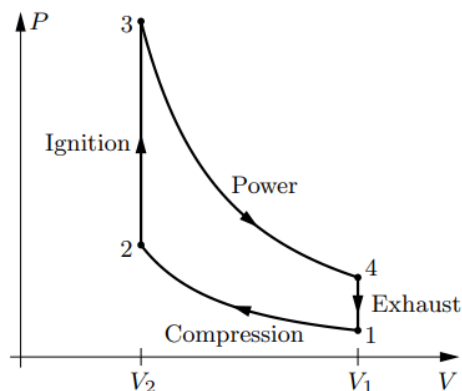


Figure 42: The idealized Otto cycle, an approximation of what happens in a gasoline engine. In real engines, the compression ratio V_1/V_2 is large than shown here, typically 8 or 10.

With an ideal gas, we can show that the efficiency is given by the form

$$e = 1 - \left(\frac{V_2}{V_1} \right)^{\gamma-1}$$

where γ is a parameter that depends on the ideal gas. For air, we have $\gamma = 7/5$ (with $f = 5$). So,

$$\frac{V_1}{V_2} = 8$$

which results in an efficiency

$$e = 1 - \left(\frac{1}{8} \right)^{2/5} \approx 56\%.$$

Practically, we have efficiencies of about 20%-30%.

3.1.3 The Steam Engine

The steam engine was discovered by Watt in 1959. Previously, steam power was used in other cases. To this day, the steam engine is still the preferred way to generate power in power plants.

The steam engine is modelled using the **Rankine cycle**. In this cycle, heat is absorbed at constant pressure in the boiler and expelled at constant pressure in the condenser, so we can write the efficiency as

$$e = 1 - \frac{Q_c}{Q_h} = 1 - \frac{H_4 - H_1}{H_3 - H_2} \approx 1 - \frac{H_4 - H_1}{H_3 - H_1}$$

where the approximation $H_2 \approx H_1$ works well in most cases since the pump adds very little energy to the water, while the PV term in H is small for liquids in comparison to gases. We can refer to the steam tables⁶ to determine the values of H . For example, suppose we have $P_{\text{low}} = 0.023$ bar, $P_{\text{high}} = 300$ bar, with $H_1 = 84$ kJ, $H_3 = 3444$ kJ, and $S_3 = 6.223$ kJ/K. From $3 \rightarrow 4$, we have that steam transitions to water plus steam at 20°C , so we have

$$S_4 = S_3 = xS_1 + (1 - x)S_v$$

⁶See Schroeder 4-136.

Here, $x = 29\%$, and thus

$$H_4 = xS_1 + (1 - x) = 1824 \text{ kJ}.$$

Thus, the efficiency is

$$e = 1 - \frac{1824 - 84}{3444 - 84} = 48\%.$$

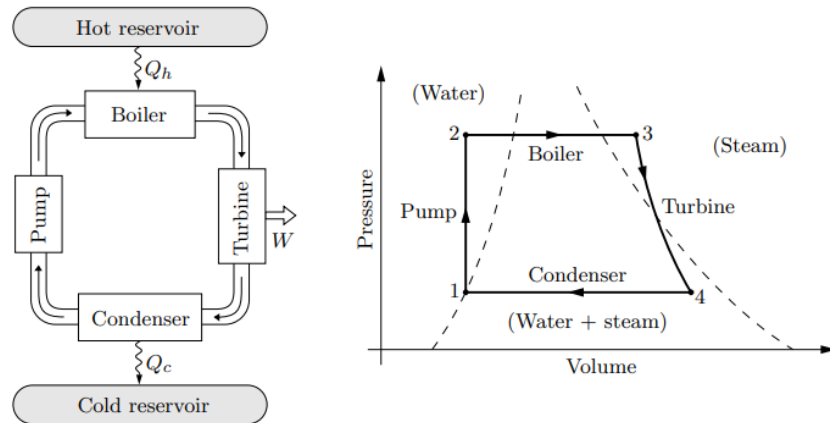


Figure 43: Schematic diagram of a steam engine and the associated PV cycle (not to scale), called the Rankine cycle. The dashed lines show where the fluid is liquid water, where it is steam, and where it is part water and part steam.

3.2 Refrigerators

A **refrigerator** is a device which can extract heat in a sustained or continuous manner. It is similar to a heat engine in that works in the reverse operation.

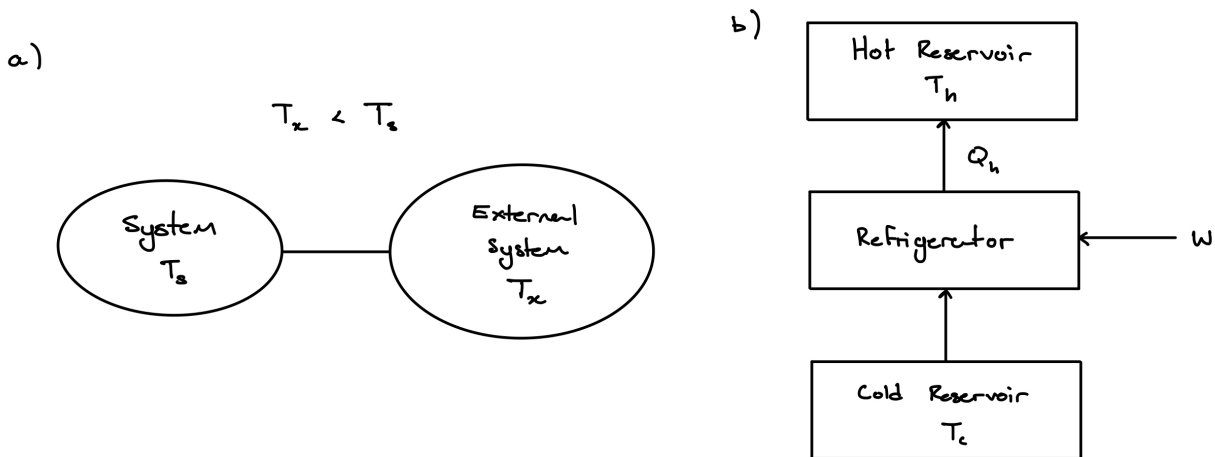


Figure 44: (a) Simple diagram representation of the temperature relation between a typical system and an external system. (b) Energy flow diagram for a refrigerator.

We define, of course, all symbols used such that $Q_h, Q_c, W > 0$ which suggests energy transfer.

In developing this working model, we shall assume that our cold (and hot) reservoirs have fixed temperatures. Indeed, there are two consequences of this assumption:

- (1) the first consequence is such that Q_c is the heat extracted from the cold reservoir, and we also have some added heat Q_a . Indeed, we will have the result

$$Q_c + Q_a = 0 \implies T_c = \text{constant}.$$

- (2) the second consequence is that we could (alternatively) assume that the temperature T_c *does* decrease, albeit very slowly.

We now ask whether we actually need a hot reservoir. Suppose, for example, we have the case in Fig. 45 with the modification that the hot reservoir is gone. Indeed, we will have that there is a net loss in work (where the work flows out of the refrigerator), and the combined system change in entropy is such that

$$\Delta S_{\text{res.}} + \Delta S_{\text{ref.}} \geq 0.$$

Indeed, we may write $\Delta S_{\text{res.}} = -Q_c/T_c$, so that we find

$$\Delta S_{\text{ref.}} \geq \frac{Q_c}{T_c} > 0.$$

That is, the entropy of the refrigerator increases constantly. Returning to the full picture, the figure of merit is the **coefficient of performance (COP)**, which we define as

$$\text{COP} = \frac{\text{benefit}}{\text{cost}} = \frac{Q_c}{W}. \quad (69)$$

From the first law of thermodynamics, we may write $Q_h = Q_c + W$, and from the second law, since the entire system is closed, we write

$$\Delta S_h + \Delta S_c + \Delta S_{\text{ref.}} \geq 0.$$

The refrigerator works in the same cycled manner as the heat engine, so over one entire cycle, we have the condition $\Delta S_{\text{ref.}} = 0$. Thus, since $\Delta S_h = Q_h/T_h$ and $\Delta S_c = -Q_c/T_c$, we find

$$\frac{Q_h}{T_h} - \frac{Q_c}{T_c} \geq 0 \implies \frac{Q_h}{Q_c} \geq \frac{T_h}{T_c}.$$

Additionally, the COP can be rewritten (since $W = Q_h - Q_c$) as

$$\text{COP} = \frac{Q_c}{Q_h - Q_c} = \frac{1}{Q_h/Q_c - 1} \leq \frac{1}{T_h/T_c - 1} = \frac{T_c}{T_h - T_c} = \text{COP}_{\text{temp.}}$$

Observe that the COP increases as $T_h - T_c$ decreases. Further, the COP can be greater than one, since there is no obvious upper limit on the quantity itself.

3.2.1 Practical Refrigerators

A typical refrigerator uses a cycle that runs as a Rankine cycle in reverse.

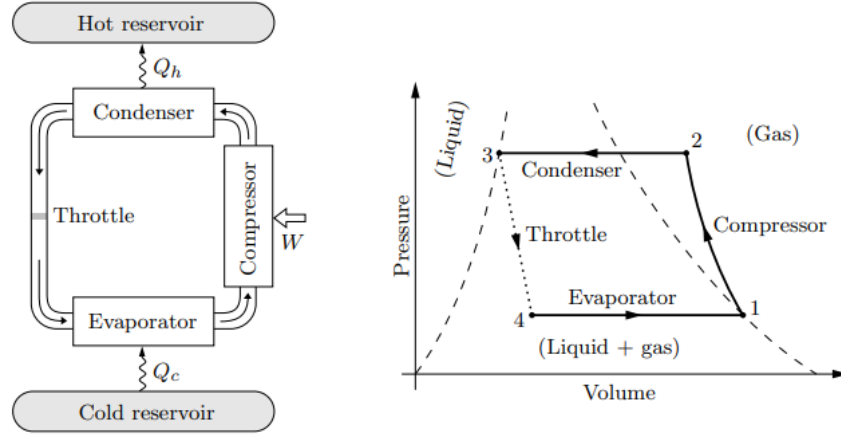


Figure 45: Schematic drawing and PV diagram (not to scale) of the standard refrigeration cycle. The dashed lines indicate where the refrigerant is liquid, gas, and a combination of the two.

Refrigerants used in practical systems include CO_2 (requires high pressure), NH_3 (toxic), Freon 12 (CCl_2F_2), which was phased out due to the impact on the Earth's ozone layer, and HFC-134a ($\text{F}_3\text{C}_2\text{FH}_2$), which was a greenhouse gas phased out in applications.

The refrigerator cycle can be summarized as follows: (1) refrigerant is in a vapour state on the brink of transition to a liquid; (1-2) adiabatic compression of the refrigerant, where work is provided by a pump; (2-3) the gas, which becomes hot after compression, dumps heat into a hot reservoir at constant pressure; (3-4) throttling process where the liquid undergoes expansion leading to its cooling; (4-1) after cooling, the refrigerant in a mix of liquid and solid, absorbs heat from a cold reservoir leading to further evaporation.

In this cycle, the coefficient of performance is such that

$$\text{COP} = \frac{Q_c}{Q_h - Q_c} = \frac{H_1 - H_4}{H_2 - H_3 + H_4 - H_1}.$$

Note that the relations $Q_h = H_2 - H_3$ and $Q_c = H_1 - H_4$ only apply since the heat is equal to the change in enthalpy for a constant pressure process.

3.2.2 The Throttling Process

The **throttling process**, or **Joule-Thompson expansion**, is an essential part of the refrigerator. The cooling of the gas is very important: how much the refrigerant cools determines how useful the refrigerator is.

A model for this process is illustrated in Figure 46. Here, the fluid is in a container with two pistons which maintain pressures P_1 and P_2 . Here, we can write

$$\Delta U = U_2 - U_1 = W_1 - W_2 = W_1 = P_1 V_1 - P_2 V_2$$

where W_1 is the work done on the system and W_2 is the work done by the system. Rearranging this, we can find that

$$U_1 + P_1 V_1 = U_2 + P_2 V_2$$

which suggests that this process takes place at constant enthalpy. Consider now an ideal gas, where we may write the

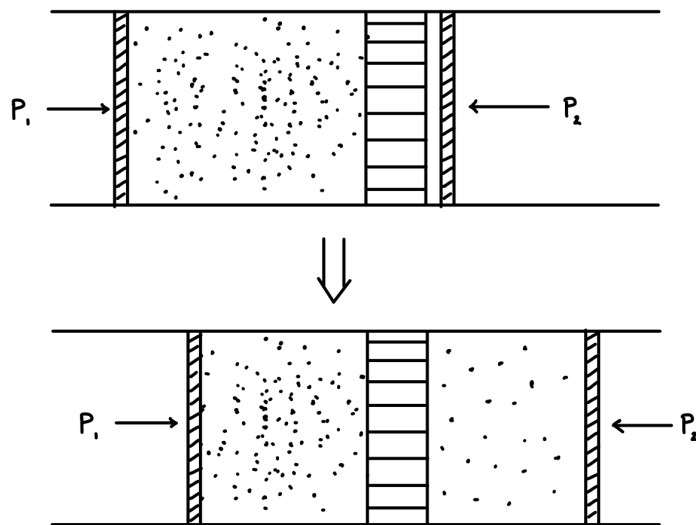


Figure 46: The two-step process of compression/expansion according to pressures P_1 and P_2 .

The enthalpy (which we did not previously define, but will), can be written as

$$H = U + PV = \frac{f}{2}NkT + NkT = \frac{f+2}{2}NkT.$$

Observe that this shows that there may be a case where H and T are constant although P varies. However, intermolecular forces play a rather important role, so we can really decompose H into

$$H = U_{\text{interaction}} + U_{\text{kinetic}} + PV$$

where, if the pressure decreases, the volume must increase leading to an increase in potential energy and a decrease in kinetic energy and temperature.

3.2.3 Refrigerator Efficiency

Using the fact that J.T. processes are isoenthalpic, we may write that $H_4 = H_3$, and the coefficient of performance becomes

$$\text{COP} = \frac{H_1 - H_3}{H_2 - H_1}.$$

Notice that the calculation of the coefficient of performance follows a method similar to that of calculating the efficiency of the steam engine. For example, we may consider data tables of a refrigerator to determine the coefficient of performance.

4 Thermodynamic Potentials

4.1 Introduction

The development of thermodynamics outlined the special role played by two state variables: the entropy S and the energy U . In this section, we discuss **thermodynamic potentials**, which are *functions of the state of a system* with the following properties:

- (1) they have dimensions of Joules;
- (2) each potential has natural variables in terms of which partial derivatives are other state variables;
- (3) mathematically, they are related to **Legendre transforms**;
- (4) an insightful physical significance emerges when considering the contact between a system and an environment/a reservoir.
 - (a) changes in thermodynamic potentials can be connected with work or heat independent from the reservoirs;
 - (b) reaching equilibrium is connected with certain properties of potentials to have a minimum or maximum.

Thermodynamic potentials are a powerful tool, essential for many applications, including phase transitions and analysis of chemical reactions.

4.1.1 Intensive and Extensive Variables in Thermodynamics

In previous sections, we made extensive use of the property of additivity for energy and entropy. Indeed, for two systems, 1 and 2, we wrote that

$$U = U_1 + U_2, \quad S = S_1 + S_2.$$

It turns out, in fact, that energy and entropy are examples of **extensive variables**.

An alternative definition of extensive variables is that of quantities which double when two (macroscopically) identical systems are placed in contact. Additive quantities, such as U and S , fulfill this definition. Similarly, the volume V and number of particles N are two obvious extensive quantities. The total magnetic moment of a system, which is obtained through summation of the macroscopic magnetic moments in a system is another example of an extensive quantity.

Naturally, we can define the concept of **intensive quantities**, which are *quantities that do not change when two identical systems are joined*. Examples of intensive quantities include the temperature T , the pressure P , and the chemical potential μ . This definition implies the following properties about our intensive quantities:

- (1) the product, division, or any algebraic combination of intensive quantities is an intensive quantity (i.e., μ/T is an intensive variable);
- (2) the product of an intensive and an extensive variable is extensive (i.e., PV is extensive since P is intensive and V is extensive);

- (3) the ratio of two extensive variables is an intensive variable (the particle density $\rho = V/N$ is intensive);
- (4) properties of derivatives follow from the above rules (i.e., consider

$$T = \left(\frac{\partial U}{\partial S} \right)_V$$

which is intensive, since U and S are extensive).

4.2 Enthalpy

We define⁷ the **enthalpy** of a system as (assuming constant pressure)

$$H = U + PV \tag{70}$$

where we note that H is derived from U by subtracting the product

$$(-P)V = \frac{\partial U}{\partial V}V.$$

4.2.1 Environment Representation

Consider now a system placed in contact with an environment through a shared movable boundary.

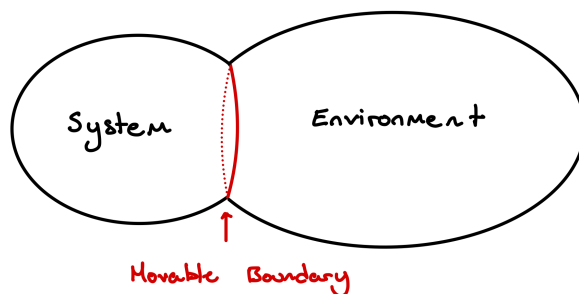


Figure 47: Coupled system and environment setup with a movable boundary separating the two.

There are a few key points to make. Notice:

- (1) the total volume is conserved. That is,

$$V + V_E = \text{const.}$$

and, the extensive variables of the same kind (U, V_E) are allowed to change while their sum remains constant;

⁷It's worth noting that enthalpy is, more generally, defined as

$$H = Q + W$$

where W is the work within a system and Q is obviously the heat in the system.

(2) the change in volume $\delta V_E = -\delta V$ is small from the perspective of the environment

$$|\delta V_E| \ll V_E.$$

However, δV may become comparable to V .

Here, all intensive variables of the environment (P_E, T_E, μ_E) are assumed to only undergo negligible changes whereas δV_E changes. In contrast, system intensive variables may undergo substantial changes.

Suppose we want to find the change in enthalpy in the isobaric case. Then, from our definition, we find

$$\Delta H = \Delta U + P\Delta V = \Delta U - W_E = Q + W_{\text{other}}$$

where we have used the first law of thermodynamics to write

$$\Delta U = Q + W_E + W_{\text{other}}.$$

Here, Q is the heat transferred to the system, W_E is the work done on the system by the environment, and W_{other} are other work components, be it electrical, chemical, etc.

One convenient physical picture is to consider enthalpy as the *energy required to build a system out of nothing **and** place it in an environment*. Indeed, U is the energy required to build that system, and PV is the work needed to "squeeze" that system into the environment. This energy can be transferred in the form of heat or other types of work.

Equivalently, the enthalpy can be seen as the total energy that one can recover if the system is annihilated (and the environment fills the created void).

4.2.2 Helmholtz Free Energy

We define the **Helmholtz free energy** as

$$F = U - TS = U - S \frac{\partial U}{\partial S} \quad (71)$$

and, consider in particular, a coupled system and environment separated by an thermally conductive wall.

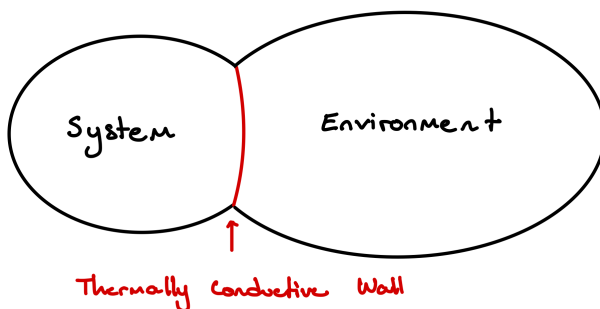


Figure 48: Coupled system and environment setup with a thermally conductive wall separating the two.

The thermally conducting wall allows for heat transfer, which we may write as

$$\delta Q = -\delta Q_E$$

where, as before, δ denotes a small change. As in the discussion of enthalpy, heat transfer does not significantly impact the environment intensive parameters. So, the change in Helmholtz free energy (assuming constant temperature in a quasistatic process) is such that

$$\Delta F = \Delta U - T\Delta S = \Delta U - Q = W.$$

When work is done on the system, a change in temperature may occur, leading to an exchange of energy with the environment. Conversely, the system can do work at a level that is given by ΔF (and not ΔU). We justify the name "free energy" by this.

One way of interpreting F is as the cost to *create a system out of nothing*. The energy required is U minus the energy TS delivered by the reservoir.

$$\Delta F = \Delta U - T\Delta S = Q + W - T\Delta S.$$

Of course, in a quasistatic process, $Q = T\Delta S$ as before, and we recover the original formulation that $\Delta F = W$. Otherwise, there is entropy production within the system, and we have the inequality condition

$$\Delta F \leq W$$

in general.

4.2.3 Gibbs Free Energy

We define the Gibbs free energy as

$$G = U - TS + PV \quad (72)$$

which has a similar definition to enthalpy, H , and the Helmholtz free energy, F . The Gibbs potential is relevant for a system that interacts with an environment that ensures a constant pressure, P and temperature, T .

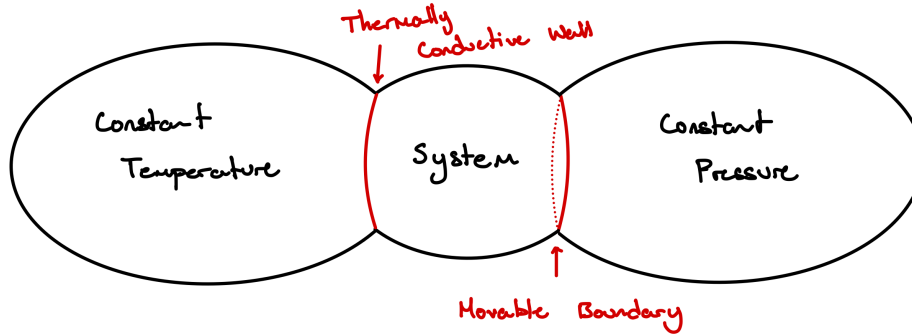


Figure 49: Coupled system and two environments inducing constant temperature and pressure, separated by a thermally conductive wall and movable boundary respectively.

Indeed, we have the small changes such that

$$\delta Q = -\delta Q_{ET}, \quad \delta W = -\delta W_{EP}$$

where E_T refers to the temperature environment, and E_P refers to the pressure environment. Assuming constant pressure and temperature, we can write the change in the Gibbs free energy as

$$\Delta G = \Delta U - T\Delta S + P\Delta V = \Delta U - Q + W = W_{\text{other}}$$

where we have used the first law of thermodynamics in the form

$$\Delta U = Q + W + W_{\text{other}},$$

and implicitly assumed that W is mechanical work. W_{other} includes work other than mechanical.

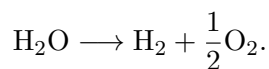
Similar to the derivation before, we can find that in a general process, we have the inequality condition

$$\Delta G \leq W_{\text{other}}.$$

4.3 Chemical Reactions

4.3.1 Electrolysis of Water

Consider the reaction given by



A system undergoing a reaction can be described using thermal physics. Transformations between different chemical compounds involve a change in energy. For example, in the reaction above, the binding energy of two H_2O molecules must be compared with that of two H_2 molecules and one O_2 molecule. This energy difference can, in principle, be calculated. However, such calculations can be rather challenging. Moreover, molecular energies by themselves are insufficient to give a useful measure of a reaction balance. This is because one must know, in general, the relevant thermodynamic potentials, whose minimization informs the balance of energies in a reaction.

Chemical reactions are analyzed using tables specifying quantities such as H , S , or G . For water, we specifically have that $\Delta H = 286$ kJ. In part, this energy is used to "squeeze" the environment that maintains a pressure ($P = 1$ atm is assumed conventionally). For each reactant, the volume associated with it is given by

$$V = \frac{m}{\rho_m} = \frac{nA}{\rho_m}$$

where n is the number of moles, A is the atomic mass in grams, and ρ_m is the mass density. We compute the volume of each molecule individually to find that

$$\begin{aligned} V_{\text{H}_2\text{O}} &= \frac{1 \text{ mol} \cdot 18.015 \text{ g mol}^{-1}}{0.997 \text{ g cm}^{-3}} = 18.069 \text{ cm}^3 \\ V_{\text{H}_2} &= \frac{1 \text{ mol} \cdot 2.016 \text{ g mol}^{-1}}{0.081 \cdot 10^{-3} \text{ g cm}^{-3}} = 24.9 \cdot 10^3 \text{ cm}^3 \\ V_{\text{O}_2} &= \frac{1 \text{ mol} \cdot 32 \text{ g mol}^{-1}}{1.29 \cdot 10^{-3} \text{ g cm}^{-3}} = 24.8 \cdot 10^3 \text{ cm}^3. \end{aligned}$$

Thus, we compute the change in volume to be

$$\Delta V = V_{\text{H}_2} + \frac{1}{2}V_{\text{O}_2} - V_{\text{H}_2\text{O}} = 24.9 \text{ L} + \frac{1}{2}24.8 \text{ L} - 0.018 \text{ L} = 37.3 \text{ L}.$$

Indeed, we find that the work done is such that

$$P\Delta V = 10^5 \text{ Pa} \cdot 37.3 \cdot 10^{-3} \text{ m}^3 = 3.7 \text{ kJ}.$$

Of the enthalpy change $\Delta H = 286 \text{ kJ}$, the amount $P\Delta V = 3.7 \text{ kJ}$ is used to change the state of the environment and 282 kJ is the energy cost.

During the reaction, the entropy increases by an amount

$$\Delta S = S_{\text{H}_2} + \frac{1}{2}S_{\text{O}_2} - S_{\text{H}_2\text{O}} = 131 \text{ JK}^{-1} + \frac{1}{2}205 \text{ JK}^{-1} - 70 \text{ JK}^{-1} = 163 \text{ JK}^{-1}$$

for a single mole. However, this change in entropy requires a heat of

$$Q = T\Delta S = 298 \text{ K} \cdot 163 \text{ JK}^{-1} = 49 \text{ kJ}.$$

We can also calculate the change in the Gibbs potential,

$$\Delta G = \Delta H - T\Delta S = 286 \text{ kJ} - (298 \text{ K})(163 \text{ JK}^{-1}) = 237 \text{ kJ}.$$

Often values of G are provided in tables as well.

References